

To Whom It May Concern,

I am pleased to share with you my recent research paper on the Fusion Energy Industry. The analysis is intended to cut through the noise surrounding fusion by providing a clear-eyed view of the scientific pathways, commercialization timelines, capital requirements, and policy frameworks shaping this burgeoning sector.

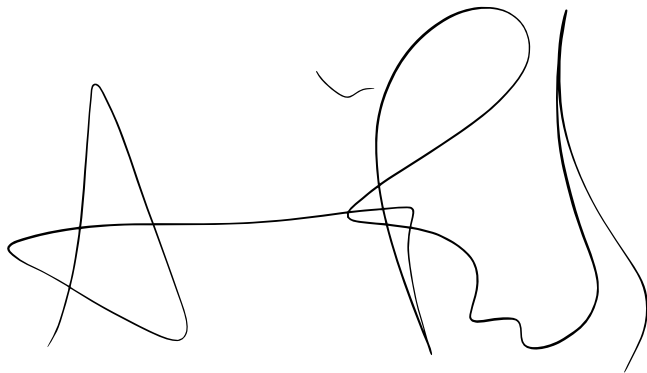
Fusion sits at the intersection of frontier science and global necessity. For decades, it has been cast as both the "holy grail" of clean energy and one of the most capital-intensive technical challenges on Earth. Yet over the last five years, real inflection points have emerged—advances in high-temperature superconducting magnets, high-energy lasers, and material sciences—that have shifted the dialogue from “if” to “when and how.”

This paper is my effort to frame how investors, policymakers, and industry leaders should evaluate fusion companies—not by hype cycles or grand promises, but by technical approach, CAPEX intensity, levelized cost of energy implications, and the efficiency of pathways to commercialization.

I hope this research contributes meaningfully to the broader dialogue and helps guide smart capital formation in a space where billions will be invested, but only a handful of approaches may ultimately scale. Please don't hesitate to reach out if you'd like to discuss the findings further.

Sincerely,

Abhi Prakash

A handwritten signature in black ink, consisting of a stylized 'A' followed by a series of loops and a long vertical stroke, representing the name Abhi Prakash.

Nuclear Fusion Industry Analysis Paper

Table of Contents

Executive Summary	3
1. Industry Landscape	4
1.1 Overview & Market Size	4
1.2. Structure of the Industry	6
1.3 Role in Climate-Change Mitigation	8
2. Technology Pathways & Approaches to Fusion	13
2.1 Tokamak & High-Field HTS Magnets	13
2.2 Pulsed-Magnet Fusion & Compact FRCs	14
2.3 Magnetized-Target Fusion (MTF)	15
2.4 Laser-Driven Inertial Confinement (ICF)	17
2.5 Z-Pinch & Sheared-Flow Plasma	19
2.6 Hybrid & Novel Driver Concepts	20
3. Enabling Systems & Economics	23
3.1. Fuel Cycle & Tritium Management	23
3.2 Critical Materials & Supply-Chain Constraints	27
3.3 Balance-of-Plant & Cogeneration Opportunities	28
3.4 Commercialization Roadmaps	29
3.5 Economics & Financing Structures	30
4. Market Opportunities & Value Creation	31
4.1 Early-Market Applications and End-Use Cases	31
4.2 Investment Opportunities	36
4.3 Key Value Drivers	39
A. Headwinds	41
B. Tailwinds	41
5. Strategic Fit & Impact	45
5.1 Fit with Investment Portfolio	45
5.2 Environmental and Climate Impact	46

5.3 Risk Landscape	47
6. Conclusion and Recommendations	48
References	50

Executive Summary

Private investment in the fusion industry has surged, with more than US \$7.1 billion raised globally as of 2024, up approximately US \$900 million from the prior year, while public funding rose 57% to about US \$426 million.¹ This influx underscores the maturation of the sector and its transition from concept to commercialization. The fusion landscape now spans more than 40 active ventures, with several pursuing distinct technological pathways including high-temperature superconducting (HTS) tokamaks, pulsed magnet-based systems, magnetized-target fusion, laser-driven inertial confinement, and compact Z-pinch reactors.

Key developments and milestones include:

- Helion Energy’s landmark PPA with Microsoft represents the world’s first agreement of its kind. The company anticipates delivering ≥ 50 MW of fusion-generated electricity by 2028, following a one-year ramp-up phase, with Constellation as the power marketer.²
- Regulatory innovation: The U.S. Nuclear Regulatory Commission (NRC) has established a lighter licensing framework for fusion under 10 CFR Part 30³, aligning with byproduct-material regulatory models rather than the more burdensome fission-centric rules. This streamlines the licensing pathway and lowers compliance hurdles.
- Technological trajectories: For instance, Commonwealth Fusion Systems (CFS) is advancing its HTS tokamak roadmap, targeting first net-energy output by 2027 with SPARC, and aiming for grid-connected power plants in the early 2030s.⁴

Together, these developments reflect critical shifts across three dimensions:

¹ Fusion Industry Association. (2024). *2024 global fusion industry report*. <https://fusionindustryassociation.org>

² Helion Energy. (2023, May 10). Helion announces world’s first fusion PPA with Microsoft. <https://www.helionenergy.com>

³ U.S. Nuclear Regulatory Commission. (2023, February). *Options for licensing and regulating fusion energy systems (10 CFR Part 30 pathway)*. <https://www.nrc.gov>

⁴ MIT Energy Initiative. (2024, September). *The role of fusion in a decarbonized electricity system*. <https://energy.mit.edu>

1. Economic: The escalating capital via both private and public channels, including targeted subsidies and milestone-based programs, has catalyzed multiple commercial-scale fusion projects.
2. Regulatory: Fast-tracked licensing provides clarity and reduces risk for developers exploring innovative reactor designs.
3. Technological: Diverse reactor architectures are advancing in parallel, boosting the overall robustness of the ecosystem rather than placing all bets on a single pathway.

The executive summary highlights that fusion is no longer an abstract aspiration. Supported by accelerating investment, regulatory accommodation, and concrete development timelines, fusion energy is emerging as a credible candidate for firm, zero-carbon power in the 2030s. Strategic investment in enabling technologies such as high-temperature superconductor magnets, tritium handling systems, and advanced control software can position stakeholders to capture attractive returns as fusion transitions to bankable infrastructure.⁵

Near-term next steps are to secure positions in critical supply chains, including high-temperature superconductor tape, lithium-6 enrichment, and high-power laser diodes, through offtake agreements or strategic equity,⁶ complete technical diligence on device performance and reliability across Q, repetition rates, component lifetimes, and closed tritium loops⁷; stress-test project economics under downside policy scenarios, including potential changes to \$45Y/\$48E credits and U.K. Contracts for Difference parameters⁸; engage anchor off-takers in data centers, hydrogen, and heavy industry to structure long-tenor PPAs and heat-offtake contracts that support project finance⁹; prioritize brownfield or national-lab sites with existing interconnections and clear pathways to Part 30 licensing while assessing community sentiment¹⁰; and define staged exit routes across IPO, M&A, or yield-oriented holds that align with milestone-based risk reduction.

1. Industry Landscape

1.1 Overview & Market Size

Nuclear fusion seeks to generate energy by fusing light isotopes of hydrogen, primarily deuterium and tritium, under extreme temperature and pressure conditions, releasing high-grade heat and electricity without direct greenhouse gas emissions. The industry has evolved

⁵ Clean Air Task Force. (2025, April). *Fusion on the grid – A U.S. fusion policy report*. <https://www.catf.us>

⁶ MIT Energy Initiative. (2024, September). *The role of fusion in a decarbonized electricity system*. <https://energy.mit.edu>

⁷ International Atomic Energy Agency. (n.d.). Fusion energy.

⁸ UK Government. (2025, June 12). *£2.5 billion for world-first prototype fusion energy plant*. <https://www.gov.uk>

⁹ Helion Energy. (2023, May 10). *Helion announces world's first fusion PPA with Microsoft*. <https://www.helionenergy.com>

¹⁰ U.S. Nuclear Regulatory Commission. (2023, February). *Options for licensing and regulating fusion energy systems (10 CFR Part 30 pathway)*. <https://www.nrc.gov>

from predominantly government-led research to a dynamic ecosystem where public and public and private sectors collaborate, with private capital now exceeding US \$7.1 billion across more than fifty companies.¹¹, supplemented by major public programs.

While fusion research is global, the United States and the United Kingdom have emerged as the most significant hubs for private investment and prototype deployment. In the U.S., developers benefit from the Department of Energy's milestone-based funding program, Inflation Reduction Act tax credits (\$45Y and \$48E), and a streamlined licensing framework under 10 CFR Part 30.^{12,13} In the U.K., the £2.5 billion STEP project aims to leverage domestic manufacturing capacity and secure a first-mover advantage in the fusion supply chain.¹⁴ Large public programs remain critical outside these markets, such as ITER in France, CFETR in China, and JT-60SA in Japan, which provide foundational scientific data for commercial-scale reactor designs.¹⁵

The present investment ecosystem spans the entire value chain:

- Core reactor developers include CFS (HTS tokamak), Helion Energy (pulsed magnet FRC), TAE Technologies, General Fusion (magnetized-target), Zap Energy (Z-pinch), and Xcimer Energy (laser-driven inertial confinement).¹⁶
- Enabling technology suppliers to produce high-temperature superconductor (HTS) wire, cryogenic systems, breeding blanket materials, precision components, and advanced plasma-control software.¹⁷
- Project finance is beginning to emerge for first-of-a-kind (FOAK) plants, with corporate off-takers like Microsoft signing long-term power purchase agreements (PPAs) that serve as templates for future deals.¹⁸

Corporate demand for fusion power is concentrated in sectors requiring high reliability and low carbon intensity. Data centers, green hydrogen producers, and heavy industry (e.g., steelmaking) see fusion as a hedge against fossil fuel volatility and tightening emissions

¹¹ Fusion Industry Association. (2024). *2024 global fusion industry report*. <https://fusionindustryassociation.org>

¹² Resources for the Future. (2025, March). *Projected impacts of repealing \$45Y and \$48E technology-neutral clean electricity tax credits*. <https://www.rff.org>

¹³ U.S. Nuclear Regulatory Commission. (2023, February). *Options for licensing and regulating fusion energy systems (10 CFR Part 30 pathway)*. <https://www.nrc.gov>

¹⁴ UK Government. (2025, June 12). *£2.5 billion for world-first prototype fusion energy plant*. <https://www.gov.uk>

¹⁵ European Commission. (n.d.). *ITER funding factsheet: EU contribution 2021–2027*. <https://energy.ec.europa.eu>

¹⁶ FIA, 2024

¹⁷ FIA, 2024

¹⁸ Helion Energy, 2023

regulations. These industries value fusion's capacity factors above 90 %, high-temperature output (>600 °C), and potential for on-site fuel self-sufficiency through tritium breeding.^{19 20}

From an industry structure standpoint, capital remains concentrated among the top five private developers, but technology diversity is high.²¹ This duality creates both resilience and competitive pressure. As FOAK demonstration plants approach commissioning, upstream suppliers, such as HTS manufacturers and tritium-handling specialists, are entering long-term supply agreements to lock in future capacity. Early vertical integration strategies by developers, including proprietary magnet manufacturing and in-house fuel cycle systems, indicate a maturing commercial mindset.

The fusion industry's trajectory over the next decade is expected to be shaped by:

1. Technology validation of net-energy operation in sustained or high-frequency regimes.²²
2. Supply chain scaling, particularly for HTS tape and tritium-breeding components.
3. Regulatory certainty, with continued policy support to reduce FOAK risk²³
4. Commercial model refinement, including modular deployment strategies and multiproduct business models combining electricity, process heat, and hydrogen production.^{24 25}

In summary, fusion's transition from scientific research to pre-commercial engineering is underpinned by strong private and public investment, supportive regulatory frameworks in lead markets, and increasing corporate interest in long-term clean energy procurement. These conditions position fusion as a credible and strategic contributor to mid-century decarbonization and energy security targets.

1.2. Structure of the Industry

The nuclear fusion sector consists of a core group of approximately fifty reactor development companies, supported by a broader network of component suppliers, research institutions, and emerging project-finance entities. Although the range of technologies under development is diverse, including tokamaks, stellarators, Z-pinch systems, magnetized-target fusion, and laser-driven inertial confinement, capital allocation remains heavily concentrated. A small number of

¹⁹ MIT Energy Initiative. (2024, September). *The role of fusion in a decarbonized electricity system*.

<https://energy.mit.edu>

²⁰ Clean Air Task Force. (2025, April). *Fusion on the grid – A U.S. fusion policy report*. <https://www.catf.us>

²¹ FIA, 2024

²² MIT, 2024; CATF, 2025

²³ MIT, 2024; CATF, 2025

²⁴ Clean Air Task Force. (2025, April). *Fusion on the grid – A U.S. fusion policy report*. <https://www.catf.us>

²⁵ MIT Energy Initiative. (2024, September). *The role of fusion in a decarbonized electricity system*.

<https://energy.mit.edu>

companies, notably Commonwealth Fusion Systems (CFS), Helion Energy, TAE Technologies, General Fusion, and Zap Energy, collectively attract the majority of private investment.²⁶

Upstream supply chains are still fragmented, with critical materials and manufacturing capabilities concentrated in limited markets. These include high-temperature superconductor (HTS) wire manufacturers, cryogenic equipment vendors, and tritium-breeding system developers. Because the scale-up of these enabling technologies is essential for timely deployment, several developers have begun pursuing vertical integration. For example, long-term offtake agreements for HTS tape are being signed to secure supply for future first-of-a-kind (FOAK) plants.²⁷

As reactor concepts progress toward commercial readiness, midstream partnerships are forming between developers, engineering-procurement-construction (EPC) contractors, utilities, and industrial off-takers. The 2023 power purchase agreement between Helion Energy and Microsoft has served as a model, demonstrating that large corporate customers are willing to commit to multi-year contracts for fusion power before plant commissioning.²⁸ This approach is beginning to be replicated by utilities and heavy industrial operators seeking long-term, zero-carbon energy sources.

The capital stack is diversifying alongside technology maturation. Early-stage financing remains dominated by venture capital and strategic corporate investments, such as Breakthrough Energy Ventures' backing of CFS and Zap Energy. Climate-focused funds like Congruent Ventures have also supported smaller-scale and novel reactor concepts, including micro-fusion systems from Avalanche Energy.²⁹ More recently, early project-finance models are being explored by infrastructure investors, with interest increasing as regulatory clarity and technology readiness levels improve.

Looking ahead, the sector is expected to experience consolidation as winning reactor architectures emerge from demonstration programs. Supply chains for critical components, especially HTS magnets and tritium systems, will need rapid capacity expansion, creating opportunities for strategic mergers or alliances. Partnerships between utilities and developers are likely to mirror patterns from other clean-energy technologies, such as offshore wind, where collaborative agreements supported accelerated deployment once early technical risks were resolved.³⁰

²⁶ Fusion Industry Association. (2024). 2024 global fusion industry report. <https://fusionindustryassociation.org>

²⁷ MIT Energy Initiative. (2024, September). *The role of fusion in a decarbonized electricity system*. <https://energy.mit.edu>

²⁸ Helion Energy. (2023, May 10). *Helion announces world's first fusion PPA with Microsoft*. <https://www.helionenergy.com>

²⁹ Congruent Ventures. (n.d.). *Avalanche Energy*. <https://www.congruentvc.com>

³⁰ Clean Air Task Force. (2025, April). *Fusion on the grid – A U.S. fusion policy report*. <https://www.catf.us>

In summary, the structure of the fusion industry is characterized by concentrated capital among a small set of leading developers, a fragmented but strategically vital supplier base, and increasing collaboration between technology developers and end users. As FOAK projects advance toward grid connection, the balance between competition and cooperation will determine both the speed and scale of commercialization.

1.3 Role in Climate-Change Mitigation

Nuclear fusion is not a singular solution to global decarbonization but offers substantial potential to reduce greenhouse gas emissions and provide firm, zero-carbon power at scale.³¹ Its primary value lies in complementing variable renewable energy sources such as wind and solar, supplying dispatchable electricity and high-grade heat without the long-lived radioactive waste associated with fission.

Studies from the MIT Energy Initiative estimate that if commercially viable fusion plants are available by mid-century, they could reduce global decarbonization costs by US\$3.6–8.7 trillion and supply up to half of the world’s electricity by 2100, depending on cost trajectories.³² Lifecycle assessments place fusion’s carbon intensity below 5 g CO₂-e/kWh, a level comparable to wind and lower than fission when fuel-cycle impacts are considered. These ultra-low lifecycle emissions allow fusion to meet even stringent regulatory thresholds for zero-carbon classification.

Fusion’s high-capacity factors, modelled at more than 90%, and outlet temperatures exceeding 600 °C enable it to decarbonize hard-to-abate sectors. Examples include:

- Green hydrogen production through solid-oxide electrolysis, which requires high-grade heat and can operate continuously for maximum efficiency.
- Synthetic fuel synthesis for aviation and shipping, which demands both high heat and constant energy input.
- High-temperature industrial processes such as steelmaking, cement production, and chemical manufacturing, where renewable electrification alone is technically challenging.

Fuel security is another climate-relevant advantage. Fusion’s primary fuel components, deuterium extracted from seawater and tritium bred on-site, are effectively inexhaustible on human timescales, removing exposure to volatile fossil fuel markets and associated emissions from extraction and transport.³³

³¹ Clean Air Task Force. (2025, April). *Fusion on the grid – A U.S. fusion policy report*. <https://www.catf.us>

³² MIT Energy Initiative. (2024, September). *The role of fusion in a decarbonized electricity system*. <https://energy.mit.edu>

³³ MIT, 2024

From a systems integration perspective, fusion’s compact footprint and ability to site plants on brownfield locations reduce the land-use impacts often cited as constraints for renewable expansion. Moreover, its steady generation profile alleviates the need for large-scale, long-duration energy storage, further reducing system-wide costs and material demand.³⁴

Early commercial momentum is being reinforced by favorable policy frameworks, robust carbon markets, and corporate procurement of 24/7 carbon-free energy. Long-term power purchase agreements, such as the Helion–Microsoft deal, demonstrate that private-sector off-takers are willing to commit to fusion power well in advance of operational deployment, accelerating the learning curve and shortening the time to market.³⁵

In sum, nuclear fusion’s role in climate change mitigation is defined by its ability to deliver constant, dispatchable, and ultra-low-emission power and heat, support the decarbonization of industrial sectors, and complement renewable energy in a balanced and resilient zero-carbon grid. If cost and deployment milestones are achieved on schedule, fusion could become a keystone technology in achieving deep decarbonization goals while enhancing energy security.

Recent Inflection Points

Between late 2021 and mid-2025, several developments shortened credible commercialization timelines and shifted attention from basic science to pre-commercial engineering.

Scientific breakeven at NIF, Lawrence Livermore National Laboratory’s National Ignition Facility, reported the first experimental shot to exceed scientific breakeven, producing 3.15 MJ of fusion output from 2.05 MJ of laser input on December 5, 2022. Subsequent shots in 2023 repeated ignition at lower input energies, confirming that target physics and symmetry can cross the ignition threshold, although large engineering gaps remain for power production.³⁶

High-field HTS magnet demonstration (September 2021, with follow-on validation in 2024). Commonwealth Fusion Systems and MIT demonstrated a large-bore 20-tesla high-temperature superconducting magnet, a scale and field strength that enables much smaller tokamak volumes at a given confinement target. MIT reported additional testing in 2024 that supports the readiness of HTS magnets for fusion-relevant operating conditions. These results underpin compact-tokamak roadmaps such as SPARC and ARC.³⁷

³⁴ CATF, 2025

³⁵ Helion Energy, 2023

³⁶ Lawrence Livermore National Laboratory. (2022, December 13). National Ignition Facility: Fusion ignition achieved. <https://www.llnl.gov>

³⁷ Commonwealth Fusion Systems. (2021, November 30). CFS closes US \$1.8 billion Series B. <https://cfs.energy> ; MIT Energy Initiative. (2024, September). The role of fusion in a decarbonized electricity system. <https://energy.mit.edu>

Policy and regulatory clarity in the United States (2024–2025). The U.S. Nuclear Regulatory Commission established a fusion-specific licensing path using the 10 CFR Part 30 byproduct material framework, distinct from fission power reactor rules. Congress reinforced this approach through the ADVANCE Act of 2024, which directs the NRC to maintain a separate lane for fusion. This change reduces licensing time and compliance costs for first-of-a-kind plants. In parallel, the Department of Energy launched a milestone-based program that committed an initial US\$46 million to eight private teams to deliver pilot-plant designs, and announced FIRE Collaboratives with up to US\$180 million available to accelerate tritium, materials, and control research.³⁸ Treasury guidance has also confirmed the applicability of the Inflation Reduction Act’s tech-neutral §45Y and §48E clean-electricity credits to fusion facilities, an economic lever referenced in sector analyses.³⁹

First long-dated corporate offtake for fusion power (May 2023). Helion Energy signed a power purchase agreement with Microsoft for at least 50 MW, targeting first delivery in 2028 with Constellation as the power marketer. The agreement provided a template for corporate procurement and signaled willingness by investment-grade buyers to contract pre-commissioning capacity when milestones are explicit.⁴⁰

Sovereign competition and program scale-up in the U.K. and E.U. (2021–2025). The United Kingdom committed £2.5 billion to the STEP prototype at West Burton A, with government statements in June 2025 positioning STEP as the anchor of a national supply-chain strategy. The European Union continued its ITER funding through the 2021–2027 budget at approximately €5.6 billion, sustaining the public research base that informs DEMO-scale engineering.⁴¹

Capital inflows and industry breadth (2024–2025). The Fusion Industry Association’s 2024 survey reported more than US\$7.1 billion of cumulative private investment, with about US\$900 million added over the prior year and public funding up 57 percent to US\$426 million. These data capture both mega-rounds for leading developers and early checks for new concepts such as long-pulse laser drivers and micro-fusion systems.⁴²

Taken together, ignition demonstrations, HTS-magnet readiness, a lighter U.S. licensing framework, milestone-based federal programs, and the first corporate PPA materially strengthen the pre-commercial case. Concurrent state and sovereign commitments, along with documented private capital growth, indicate that the next gating items are sustained $Q > 1$ operation in power-plant-relevant devices, tritium breeding at throughput, and component

³⁸ NRC, 2023; RFF, 2025

³⁹ RFF, 2025

⁴⁰ Helion Energy, 2023

⁴¹ European Commission, n.d.

⁴² Fusion Industry Association, 2024 *global fusion industry report* (2024).

lifetime under neutron load. These milestones now define the near-term basis for due diligence and investment sequencing.^{43 44}

Private-Capital Surge

Private investment in nuclear fusion has accelerated sharply over the past three years, driven by technical milestones, policy clarity, and the emergence of credible commercial timelines. According to the Fusion Industry Association’s 2024 Global Fusion Industry Report, cumulative private-sector funding reached US \$7.1 billion by mid-2024, up from approximately US \$6.2 billion in 2023.⁴⁵ Of this total, roughly US \$900 million was raised in the preceding twelve months, indicating continued investor confidence despite broader capital market volatility.

The distribution of this funding reflects a concentration toward both high-profile “mega-rounds” and targeted early-stage financing. At the upper end, Commonwealth Fusion Systems’ US\$1.8 billion Series B and Helion Energy’s multi-hundred-million-dollar raises have provided the capital base for first-of-a-kind plant construction and component manufacturing scale-up.⁴⁶ In parallel, venture firms such as Breakthrough Energy Ventures have allocated significant follow-on capital to mature technology developers. At the same time, Congruent Ventures and other early-stage funds have backed smaller, novel concepts such as micro-fusion reactors from Avalanche Energy.⁴⁷

The investor profile has broadened beyond traditional venture capital. Climate-focused mega-funds, strategic corporates, and technology infrastructure investors are increasingly present in funding rounds, reflecting a shift from speculative R&D plays toward infrastructure-scale assets. This diversification supports a more complete capital stack, enabling developers to address both technology risks and early project-finance requirements.

Public funding has also moved in parallel with private inflows. FIA data show government fusion funding climbing 57% year-on-year to more than US \$400 million in 2024, with allocations to demonstration programs, milestone-based grants, and enabling research.⁴⁸ While small relative to private capital totals, this public funding provides critical non-dilutive support for high-risk subsystems such as tritium-breeding technology, high-temperature materials, and plasma-control software.

Importantly, early-stage project-finance models are beginning to emerge. The Helion–Microsoft 50 MW power purchase agreement (PPA), scheduled for first delivery in 2028, provides an early

⁴³ MIT, 2024

⁴⁴ CATF, 2025

⁴⁵ FIA, 2024

⁴⁶ CV, n.d.

⁴⁷ CV, n.d.

⁴⁸ FIA, 2024

template for bankable offtake arrangements in the fusion sector.⁴⁹ Similar PPA frameworks are now being explored by other developers with utilities and heavy-industry buyers, with the potential to accelerate private debt participation once technical and licensing milestones are achieved.

In summary, the private-capital surge in fusion is characterized by record cumulative funding, a diversification of investor types, alignment with supportive public programs, and the emergence of long-term offtake contracts that bridge the gap between venture-style risk capital and infrastructure-grade finance. This convergence is critical for sustaining momentum toward the first wave of commercially relevant plants.

Global Market Size

Independent market analysts estimate that commercial activity tied to fusion could reach substantial scale by the end of this decade. Research and Markets values the global fusion opportunity at about US \$348.2 billion in 2024, projecting growth to roughly US \$490.0 billion by 2030, which implies a compound annual growth rate of 5.7% over the period. These figures primarily reflect revenue associated with electricity provision from fusion and do not fully account for adjacent value streams such as industrial heat, hydrogen and ammonia production, or water desalination that may be co-sited with first-of-a-kind plants, indicating possible upside as multiproduct business models mature.

Context from the investment side supports the plausibility of a multi-hundred-billion-dollar addressable market. The Fusion Industry Association's 2024 survey reported approximately US \$7.1 billion in cumulative private funding and an additional US \$900 million raised in the preceding year, alongside a 57% year-over-year rise in public funding to about US \$426 million. While these capital flows are small relative to the overall market size projections, they signal growing investor confidence and help finance the supply-chain and demonstration steps required to translate market potential into contracted capacity.

In practical terms, market realization will depend on three pacing items that are embedded in most commercial forecasts: timely validation of plant-relevant performance, the scale-up of critical materials and components, and the availability of long-tenor offtake contracts that support project finance. If developers achieve early cost targets and secure anchor buyers in power-intensive sectors such as data centers and hydrogen hubs, the 2030-level projections focused on electricity sales may understate the total monetizable output once cogeneration is included.

⁴⁹ Helion Energy, 2023

2. Technology Pathways & Approaches to Fusion

2.1 Tokamak & High-Field HTS Magnets

Tokamaks confine plasma in a toroidal chamber using a combination of toroidal and poloidal magnetic fields, heating a deuterium–tritium fuel mix to temperatures exceeding 100 million °C until fusion occurs. Traditional designs, such as JET and ITER, rely on low-temperature superconductors (NbTi/Nb₃Sn) that limit magnetic field strength to around 12 teslas. This constraint forces reactors to be extremely large, often the size of stadiums, to achieve the confinement times required for net-energy gain.

Commonwealth Fusion Systems (CFS) and the Massachusetts Institute of Technology have addressed this scaling limitation through the use of high-temperature superconductor (HTS) tape made from REBCO (rare-earth barium copper oxide). In 2021, the team demonstrated a 20-tesla, large-bore HTS magnet, representing the strongest of its kind and offering roughly 100 times greater energy density compared to conventional fusion magnets.⁵⁰ The higher magnetic field compresses the plasma more effectively, enabling smaller, more cost-efficient devices. CFS's SPARC design reduces the major radius to 3.3 meters, approximately one-fortieth the volume of ITER, while maintaining the necessary confinement parameters.

Commercial Roadmap

SPARC is under construction in Massachusetts, with first plasma targeted for 2026 and sustained net-energy operation targeted after first plasma. Following a successful SPARC demonstration, CFS plans to deploy its first commercial ARC plant, a 400 MW electric system in Virginia, in the early 2030s, financed in part by the US\$1.8 billion Series B round completed in 2021.⁵¹ MIT projections estimate first-of-a-kind (FOAK) capital expenditure at US \$2–3 billion per gigawatt and a long-term levelized cost of energy (LCOE) target of US \$40–60/MWh, competitive with combined-cycle natural gas with carbon capture.⁵²

Engineering Challenges

- HTS Tape Costs: Current REBCO tape pricing of approximately US \$100/kA-m must decrease by a factor of three to four to meet ARC cost targets.
- Neutron Flux: High-energy neutrons degrade copper stabilizers and structural steels; research is ongoing into low-activation alloys and liquid blanket configurations to improve durability.

⁵⁰ MIT, 2021

⁵¹ Commonwealth Fusion Systems. (2021, November 30). *CFS closes US \$1.8 billion Series B*. <https://cfs.energy>; Reuters. (2021, November 30). *Fusion breakthrough: CFS closes record Series B funding round*. <https://www.reuters.com>

⁵² MIT, 2024

- Thermal Integration: HTS coils operate at cryogenic temperatures (~20 K), requiring careful thermal isolation from primary coolant loops that reach 600 °C in the ARC design.

Operational Profile and Applications

High-field tokamaks can provide reliable baseload electricity through supercritical CO₂ turbines, deliver process heat for hydrogen electrolysis and ammonia synthesis, and produce fresh water through co-located desalination. Their high rotational inertia also contributes to grid frequency stability.

2.2 Pulsed-Magnet Fusion & Compact FRCs

Pulsed-magnet fusion departs from steady-state confinement in favor of rapidly formed, high-energy plasma structures. Helion Energy’s approach involves firing two field-reversed configuration (FRC) plasmoids toward each other at speeds exceeding 1.6 million km/h within a linear accelerator system. The plasmas collide and are magnetically compressed to fusion conditions in pulses of about one millisecond. This cycle repeats at approximately 1 hertz, with the same magnetic field that compresses the plasma also inducing a direct electrical current in surrounding coils, enabling direct electricity conversion at over 95 percent efficiency without a steam turbine.⁵³

Commercial Roadmap

Helion’s seventh prototype, Polaris, achieved 100-million-degree plasmas in 2023. Construction of the first commercial-scale unit, Orion, began in 2025 in Malaga, Washington, under a 50 MW PPA with Microsoft for first delivery in 2028. FOAK capital expenditure is projected in the low hundreds of millions for a 50 MW-electric module, or about US \$1 billion per gigawatt at scale.⁵⁴

Type One Energy is pursuing a hybrid stellarator-spheromak concept, combining continuous stellarator stability with pulsed heating. Early 2025 tests of its modular HTS coil set achieved target magnetic fields, and the Infinity One prototype has cleared initial design review. The company envisions factory-built 200 MW modules with LCOE targets between US \$45 and US \$65/MWh after learning curve effects.

Engineering and Cost Variables

- Pulse Electronics: Components must endure more than 30 million cycles over a decade without significant degradation.
- Plasma Stability: FRC plasmas require strong shear flows to suppress instabilities.

⁵³ Helion Energy, 2023

⁵⁴ Helion, 2023; Reuters, 2025

- Thermal Fatigue: Materials must withstand repeated high-temperature shock loads.

Operational Profile and Applications

Pulsed systems can operate as quasi-continuous baseload resources or follow demand with rapid ramping capability. Compact plant footprints make them suitable for brownfield retrofits, remote installations, and industrial sites requiring high-temperature heat (>700 °C).

2.3 Magnetized-Target Fusion (MTF)

Magnetized-target fusion combines aspects of magnetic confinement with inertial compression. A pre-heated, magnetized plasma seed is rapidly compressed by a mechanically or electromagnetically driven liner, reaching fusion-relevant densities and temperatures in milliseconds. The liner acts as both the driver and the first wall, absorbing neutrons and, in many cases, breeding tritium via lithium content.

Examples

- General Fusion uses 300 steam-driven pistons to symmetrically implode a liquid lead–lithium vortex containing the plasma.
- Liberty Fusion is a U.S. startup founded in 2025 and headquartered in New Mexico. The company is pursuing plasma-jet–driven magneto-inertial fusion, an alternative inertial approach that seeks to compress fuel using high-velocity plasma jets. Publicly available information on Liberty Fusion remains limited; as of mid-2025, the company has not disclosed verified funding rounds, pilot milestones, or commercial capacity targets.

Commercial Roadmap

General Fusion’s LM26 machine in British Columbia completed assembly in late 2024 and achieved first plasma compression in April 2025. The company targets a scientific breakeven-equivalent in 2026. In parallel, a 70 percent-scale non-electric Fusion Demonstration Plant is being built at the UKAEA Culham Campus, with planning consent granted in 2023 and commissioning expected around 2026–27. The Culham facility will not generate electricity, and the company has not published firm commercial power ratings, CAPEX, or LCOE targets. Liberty Fusion was founded in 2025 and is pursuing plasma-jet–driven magneto-inertial fusion; as of mid-2025, the company has not published verified funding rounds, milestones, or plant-capacity targets.

Key Technical Risks

- Symmetric compression. Liner collapse must stay within $\pm 3\%$ shape uniformity; piston mistiming or jet breakup can quench the burn
- Liner reusability. Liquid-metal vortices recirculate, but splashing and cavity erosion may raise maintenance costs.

- Seed-plasma lifetime. Field-reversed configurations must survive until peak compression; improved helicity injection and magnetic mirrors are under test.

Operational Profile and Applications

MTF units suit mid-scale regional baseload markets and industrial process heat applications. The modular design allows factory fabrication and rail transport, reducing site construction risk. Heat recovery between pulses enables integration with hydrogen production, ammonia synthesis, and thermal storage systems.

Physics & Engineering Approach

Feature	General Fusion	Liberty Fusion (PMFJ)
Driver	300 steam-powered pistons fire simultaneously to implode a 3-m liquid-lead-lithium vortex.	Sixteen modular plasma-magnet guns inject high- β FRCs into a central vacuum vessel; electromagnetic pistons deliver the compression pulse.
Plasma Seed	Compact torus (FRC) pre-heated to ~ 150 eV with coaxial helicity injection.	Axial FRC at ~ 200 eV, formed by PMFJ's pulsed-magnet guns.
Liner Material	Lead–lithium (Pb-Li) swirl tank; doubles as tritium breeder.	Thin liquid-metal layer plus solid metal shell for higher peak pressure.
Pulse Rate	1 Hz target $\rightarrow$ 9 GJ thermal per pulse for a 70 MW-e module.	0.5–1 Hz in the first plant; long-term goal 5 Hz (higher throughput).

Commercial Timelines & CAPEX

- General Fusion: The LM26 machine in British Columbia completed assembly in late 2024 and achieved first plasma compression in April 2025. The company is targeting a scientific breakeven-equivalent in 2026. In parallel, a 70 percent-scale non-electric Fusion Demonstration Plant is being built at the UKAEA Culham Campus, with planning consent granted in 2023 and commissioning expected around 2026–27. The Culham facility will not generate electricity, and the company has not published firm commercial power ratings, CAPEX, or LCOE targets.
- Liberty Fusion: Founded in 2025 and based in New Mexico. Public profiles describe Liberty as an early-stage venture pursuing plasma-jet–driven magneto-inertial fusion. As of mid-2025, detailed project milestones, plant capacities, funding rounds, and CAPEX or LCOE metrics have not been published.

Key Technical Risks

1. Symmetric Compression – Liner collapse must stay within ± 3 % shape uniformity; piston mis-timing or jet break-up can quench the burn.
2. Liner Re-usability – Liquid-metal vortices recirculate, but splashing and cavity erosion may raise maintenance costs.
3. Seed-Plasma Lifetime – FRCs must survive until peak compression; improved helicity injection and magnetic mirrors are under test.

Operational Implications & Applications

- Mid-scale baseload power, 70 – 150 MW blocks suit regional utilities and coal-plant brownfields.
- Process heat (500 – 650 °C), Liquid-metal coolant delivers steady heat between pulses, ideal for green-steel direct-reduced iron or ammonia synthesis.
- Rapid manufacturing & transport, Module size allows factory build and rail transport, lowering site-labor risk relative to large tokamaks.

Because MTF units run in one-second bursts, they naturally integrate with pumped-thermal or molten-salt storage, enabling flexible output without cycling the fusion core. Direct Li-Pb coolant loops simplify tritium extraction and may reduce overall balance-of-plant complexity.

Bottom line: Magnetized-target fusion offers a pragmatic cost-risk compromise: magnetic pre-heating shrinks driver energy, while mechanical or jet liners avoid the billion-dollar laser arrays of pure inertial systems. If General Fusion's Culham demo proves linear symmetry and pulse endurance, MTF could carve out the first commercial niche in 70–150 MW distributed baseload plants, complementing larger tokamak and pulsed-magnet architectures in the 2030s grid mix.

2.4 Laser-Driven Inertial Confinement (ICF)

Laser-driven inertial confinement fusion seeks to achieve net energy gain by rapidly compressing and heating a small pellet of deuterium-tritium fuel using extremely powerful laser pulses. In a typical central hot-spot design, a target capsule, often millimeters in diameter, is placed inside a cylindrical hohlraum. Multiple high-energy laser beams enter the hohlraum, striking its inner walls and producing a bath of x-rays. These x-rays symmetrically ablate the outer layer of the capsule, causing it to implode at velocities of around 300–400 km/s. The implosion compresses the fuel to densities exceeding 1,000 times that of solid hydrogen and raises its core temperature above 100 million °C, initiating fusion reactions.

Recent Achievements and Public-Sector Leadership

The most notable progress in ICF comes from the U.S. National Ignition Facility (NIF) at Lawrence Livermore National Laboratory. In December 2022, NIF achieved scientific “ignition” by producing 3.15 MJ of fusion energy output from a 2.05 MJ laser input to the target,

representing an energy gain (Q) of approximately 1.54.⁵⁵ In 2023 and 2024, NIF repeated and slightly improved on these results, demonstrating reproducibility and incremental increases in yield. While these experiments still require about 300 MJ of electrical input to power the lasers, meaning the facility operates far from net-electric gain, they represent a milestone in fusion research.

France's Laser Mégajoule facility, operated by the French Alternative Energies and Atomic Energy Commission (CEA), focuses primarily on defense-related research but also contributes to inertial fusion science. In the United Kingdom, the Central Laser Facility's Orion system supports high-energy-density plasma studies relevant to ICF target design.

Private-Sector Activity

Private entrants are developing alternative inertial fusion approaches intended to sidestep some of the limitations of large, single-shot laser systems. First Light Fusion uses a hypervelocity projectile to compress the target, while Marvel Fusion is pursuing picosecond-duration, petawatt-scale laser pulses to drive non-thermal fusion processes. These approaches aim for higher repetition rates and more compact plant designs than NIF's current configuration.

Commercialization Challenges

Key barriers to commercial-scale ICF include:

- **Laser Efficiency:** NIF's flashlamp-pumped lasers are under 1 percent wall-plug efficiency. Diode-pumped solid-state systems demonstrate about 20–35 percent in modern installations, and some specialized systems report up to roughly 50 percent, although not yet at fusion-plant scale.
- **Repetition Rate:** NIF operates at a few shots per day, whereas power plants will need 5–10 shots per second to sustain output in the hundreds of megawatts.
- **Target Fabrication:** Experimentally produced capsules are about US \$2,500 each today. Studies of automated mass production suggest a goal of about US \$0.25–0.30 per target.
- **Driver Reliability:** High-repetition-rate laser systems must maintain beam quality and alignment over billions of pulses without excessive maintenance downtime.

ICF quick facts for slides

- **Innovation highlights:** First Light Fusion (projectile-driven impact); Marvel Fusion (ultrashort, high-intensity lasers with ELI-NP partnership).

⁵⁵ MIT, 2024

- Core challenges: laser efficiency (<1 percent at NIF vs about 20–35 percent for DPSSLs), repetition rate (few shots per day vs about 5–10 Hz), capsule cost (about US \$2,500 now to about US \$0.25–0.30 goal), driver reliability over billions of shots.
- Commercial outlook: engineering breakthroughs are required before FOAK plants become viable; ambitious LCOE goals target about US \$50–80/MWh once efficiency, target cost, and repetition rate are solved.

Commercial Roadmap and Potential

The consensus in both public and private programmes is that significant engineering breakthroughs in laser technology, target fabrication, and reactor chamber design are required before ICF can approach FOAK plant deployment. A hypothetical commercial ICF plant producing 500 MW-electric at 10 Hz shot rate could have an LCOE in the range of US \$50–80/MWh if technical targets for laser efficiency, driver cost, and target supply are met.⁵⁶ Such plants could serve as high-capacity-factor, dispatchable clean power sources, as well as drivers for industrial processes requiring pulsed high-energy-density inputs.

2.5 Z-Pinch & Sheared-Flow Plasma

The Z-pinch approach to fusion uses powerful electrical currents running through a plasma column to generate its magnetic confinement field. In its simplest form, when a high current passes axially through the plasma, it produces an azimuthal magnetic field that compresses the plasma inward, or “pinches” it, toward the axis. If the compression is fast and stable enough, fusion-relevant temperatures and densities can be achieved before the plasma disassembles.

Principles and Recent Advances

Historically, Z-pinches were plagued by magnetohydrodynamic (MHD) instabilities, particularly the $m=0$ “sausage” and $m=1$ “kink” modes, which caused the plasma column to break apart before fusion conditions could be maintained. Recent developments in sheared-flow stabilization have shown promise in suppressing these instabilities. This technique introduces a controlled variation in plasma flow velocity across the column, which disrupts the growth of MHD modes and prolongs confinement times.

The most visible progress in this field comes from Zap Energy, which is developing a linear device where the plasma is formed, accelerated, and compressed in a single Z-pinch stage with sheared-flow stabilization. In 2022–2024, Zap Energy’s FuZE device demonstrated stable operation at plasma currents exceeding 650 kA, with electron temperatures approaching fusion-relevant levels. The company’s goal is to achieve $Q > 1$ in a compact, modular system that does not require superconducting magnets or large laser systems.

⁵⁶ MIT, 2024

Commercial Potential

Z-pinch reactors offer an inherently compact design, potentially reducing capital costs and enabling deployment in smaller-scale markets or industrial sites. The absence of large, cryogenic magnet systems simplifies the plant architecture and may lower operational complexity. However, commercialization timelines will depend heavily on demonstrating sustained stability at higher currents, scaling device length without introducing new instability modes, and integrating an efficient neutron-absorbing blanket for tritium breeding and energy capture.

Technical and Engineering Challenges

- **Plasma Stability:** Ensuring MHD stability at fusion-relevant densities and temperatures remains the central challenge.
- **Electrode Erosion:** High-current electrodes face erosion over repeated shots, affecting device longevity.
- **Energy Recovery:** Direct energy conversion concepts could improve efficiency but require demonstration at scale.
- **Blanket Integration:** Incorporating a tritium-breeding and energy-capture blanket without disrupting plasma stability is still unproven.

Table 1.
Summary of Z-Pinch / Sheared-Flow Plasma Approach

Parameter	Current Status (2024)	Target for Commercialization	Key Risks / Gaps
Peak Plasma Current	~650 kA (Zap Energy – FuZE)	>1 MA sustained	Stability control at higher currents
Electron Temperature	Approaching 10 keV	>15 keV	Avoiding new instability modes
Pulse Rate	Low repetition (seconds–minutes)	Continuous or high repetition (Hz)	Electrode and power supply design
Plant Architecture	Compact, no large magnets	Modular, factory-built	Integration with blankets and heat systems
Projected LCOE	Not yet estimated	US \$50–70/MWh (theoretical)	Dependent on stability and scaling success

2.6 Hybrid & Novel Driver Concepts

A growing group of fusion start-ups is pursuing cost-disruptive driver systems that bypass billion-dollar laser halls and cryogenic magnet assemblies in favor of more compact, electromechanical, and impact-driven approaches. These designs leverage technologies from

capacitor banks, railgun physics, and hypersonic materials research to create simpler, lower-cost driver architectures.

Electromagnetic Pulse-Liners and Piston Systems

Pacific Fusion’s “electromagnetic pulse-liner” concept uses thousands of modular 15-kV capacitors to deliver a 40 mega-ampere current into a thin aluminum shell. In a hundred-nanosecond event, the shell implodes at high velocity, compressing a magnetized plasma target to fusion-relevant pressures and temperatures. This design removes the need for large superconducting magnets and reduces complexity in cryogenics.

A variant of this method is being developed by Liberty Fusion, which replaces steam-powered pistons (seen in mechanical magnetized target fusion) with electromagnetic pistons. These pistons symmetrically compress a plasma-magnet-jet target, aiming for lower operating costs and higher shot rates.

Hybrid Beam-Laser Systems

Focused Energy blends short-pulse laser compression with a high-current proton beam for “fast ignition,” in which a hot spot is triggered after primary compression. This hybrid approach aims to reduce total laser energy requirements and lower costs associated with precision optics, potentially enabling more compact plants.

Hyper-Velocity Impact Systems

First Light Fusion’s two-stage gas gun launches a 100-gram flyer plate at approximately 7 km/s into a proprietary target designed to focus the resulting shockwave much like a shaped charge. Early testing has achieved multi-gigapascal pressures. The company projects that a 150 MW electric plant could be operational in the early 2030s at the cost of about US \$1.7 billion per gigawatt.

Other defense-linked laboratories are exploring hypersonic projectile systems that compress a liquid-metal target chamber in microseconds. These efforts often benefit from Department of Defense funding due to their dual-use potential.

Shared Advantages and Technical Risks

Advantages:

- No need for superconducting magnet systems or extensive optical facilities.
- Driver components can be mass-produced using conventional manufacturing (e.g., automotive or aerospace supply chains).
- Potential for mid-scale, modular deployment (100–300 MW) suitable for coal plant repurposing or grid peaking.

Major Technical Risks:

1. Driver Symmetry: Micron-scale alignment errors or microsecond-level timing mismatches can prevent ignition.
2. Target Economics: Precision targets currently cost hundreds of dollars; commercial viability requires <\$10 per target and shot rates of 1–10 Hz.
3. Component Durability: Barrels, electrodes, and liquid liners must withstand tens of millions of high-load cycles without significant degradation.
4. Wall-Plug Efficiency: To achieve a post-learning-curve LCOE in the US \$55–80/MWh range, overall driver-to-electric conversion efficiency must exceed 25 percent.

Commercialization Potential

If the late 2020s solve symmetry, target cost, and durability challenges, these hybrid driver systems could open a “fast-follower” commercialization path. Their relatively low infrastructure requirements make them attractive for repowering existing fossil plant sites and supporting grids with flexible, dispatchable clean power.

Table 2.

Summary of Hybrid & Novel Driver Concepts

Concept Type	Example Developer(s)	Driver Mechanism	Current Status	Key Challenges	Target Market Profile
Electromagnetic Pulse-Liner	Pacific Fusion	Capacitor-driven aluminum shell implosion	Prototype-stage testing	Symmetry, component lifespan, efficiency	Mid-scale, modular plants
Electromagnetic Piston	Liberty Fusion	Plasma-magnet-jet compressed by EM pistons	Prototype under development	Cost-effective targets, shot rate scalability	Grid peaking, industrial sites
Hybrid Laser-Beam (Fast Ignition)	Focused Energy	Laser compression + high-current proton beam	Lab-scale experiments	Beam–target coupling, optics cost reduction	Smaller plants, specialized loads
Hyper-Velocity Impact	First Light Fusion	2-stage gas gun impact onto a shock-focusing target	Achieved multi-GPa pressures	Barrel wear, target cost, precision impact control	Coal plant repurposing, modular
Hypersonic Projectile Compression	Defense-linked R&D labs	The projectile compresses the liquid-metal chamber	Early-stage R&D	Chamber durability, precision timing	Dual-use, government-backed

3. Enabling Systems & Economics

3.1. Fuel Cycle & Tritium Management

The primary fuel cycle for near-term fusion concepts is the deuterium–tritium (D–T) reaction, which fuses two heavy hydrogen isotopes to produce a helium nucleus, a 14.1 MeV neutron, and a total of 17.6 MeV of energy.⁵⁷

Deuterium Availability

Deuterium (^2H) is abundant and non-radioactive, constituting roughly one in every 6,420 hydrogen atoms. This equates to ~33 grams per cubic meter of seawater, yielding multi-trillion-tonne reserves globally—effectively inexhaustible on human timescales.⁵⁸ Commercially, deuterium is already available from gas suppliers at a few thousand dollars per kilogram. While its availability is not a limiting factor for fusion, scaling supply chains will be necessary for large-scale deployment.

Tritium Constraints

Tritium (^3H) is the single greatest constraint on the commercialization of deuterium–tritium fusion. Unlike deuterium, which is abundant, tritium is scarce, radioactive, and short-lived, with a half-life of just 12.3 years. Today, it is produced almost exclusively as a by-product of heavy-water fission reactors, primarily Canada’s CANDU fleet.⁵⁹ Global civilian-accessible reserves are estimated at fewer than 25 kilograms and are shrinking each year due to natural decay and reactor retirements. To put this in perspective, a single 1 GW(e) fusion power plant would require on the order of 100–130 kilograms of tritium annually—enough to consume the world’s entire inventory in its first year of operation.⁶⁰ This mismatch makes one thing clear: without reliable on-site tritium breeding, enabled by lithium-based breeding blankets and neutron multipliers, fusion cannot scale beyond the demonstration stage.

Breeding Solutions

The only viable solution to tritium scarcity is the development of breeding blankets—lithium-based layers that capture high-energy neutrons from the fusion plasma and convert lithium-6 into tritium and helium, releasing an additional 4.8 MeV of heat in the process.⁶¹ Current designs under study include molten lithium-lead (LiPb) loops and ceramic breeder pebble beds

⁵⁷ MIT, 2024

⁵⁸ MIT, 2024

⁵⁹ Kleinman Center, 2025

⁶⁰ NEI, 2023

⁶¹ NRC, 2023

such as lithium-titanate (Li_2TiO_3). To enable commercial fusion, these systems must meet three critical requirements:

- Tritium Breeding Ratio: Achieve a TBR ≥ 1.1 to offset decay and processing losses.
- Continuous Extraction: Deliver tritium at a rate sufficient to sustain plasma fueling.
- Containment: Prevent excessive tritium permeation through structural materials, ensuring compliance with strict regulatory and environmental standards.⁶²

Without such breeding systems, tritium scarcity will remain the hard stop preventing fusion from advancing beyond demonstration plants.

Comparing Breeding Blanket Approaches

Two leading tritium breeding strategies are under development, each with distinct advantages and challenges:

- Molten Lithium-Lead (LiPb) Loops
 - *Pros*: Functions as both breeder and heat-transfer medium; offers excellent neutron economy; capable of high-temperature operation, enabling more efficient power cycles.
 - *Cons*: Highly corrosive to steels, requiring advanced protective coatings; tritium extraction from liquid metal is technically demanding and not yet proven at a commercial scale.
- Ceramic Pebble Beds (Li_2TiO_3 , Li_4SiO_4)
 - *Pros*: Chemically stable and relatively well understood; modular, granular design facilitates fabrication; tritium release characteristics studied extensively in ITER test blanket programs.
 - *Cons*: Low thermal conductivity; requires separate helium coolant loops, adding system complexity; large-scale pebble fabrication and handling for DEMO plants remain unresolved.

Industry Outlook:

ITER and DEMO projects are pursuing parallel testing of both approaches through Test Blanket Modules, as no single solution has yet emerged as superior. In the long run, liquid-metal breeders may unlock higher efficiency and better integration into advanced power cycles. Ceramic pebble beds, however, are further along in development and more thoroughly characterized today. Both approaches are likely to serve as critical stepping-stones toward achieving commercial tritium self-sufficiency.

Engineering and Policy Challenges

⁶² Federation of American Scientists. (n.d.). Fusion energy development and policy insights. <https://fas.org>

Fusion's promise is immense, but turning physics into power requires solving some of the toughest engineering and governance problems humanity has faced.

Engineering Hurdles

Each gigawatt-class fusion plant will require tens of tonnes of enriched lithium-6, yet today's enrichment capacity—geared toward defense rather than energy—must scale by orders of magnitude before DEMO construction can begin.⁶³ Once operating, a plant will generate 1–3 grams of tritium per hour, which must be extracted from LiPb loops or ceramic pebbles within minutes and recycled into the plasma without bottlenecks. At the same time, tritium's permeability demands advanced containment and barrier materials to prevent leakage into structural steels and the environment. Together, these challenges define the cutting edge of fusion engineering.

Policy and Regulatory Barriers

Even with technical breakthroughs, fusion cannot scale without supportive policy frameworks. Tritium is radioactive, tightly regulated, and subject to strict release limits such as NRC 10 CFR Part 30.⁶⁴ Early commercial plants will each require several kilograms of tritium at start-up—material that is not available in surplus today. Since most existing tritium is tied to national defense programs, coordinated global stockpiling and transparent governance mechanisms will be essential by the early 2030s.⁶⁵

Bringing It Together

Fusion's success will depend not only on engineering ingenuity but also on regulatory foresight and international cooperation. Breakthroughs in the lab can build the machines, but only global trust and coordination will ensure they can be fueled and operated safely.

Fuel-Cycle Definition

The deuterium–tritium (D–T) fuel cycle is a partially closed loop: deuterium, which is abundant and externally sourced, is paired with tritium, which must be regenerated internally. The cycle proceeds through four interconnected stages. First, deuterium and a start-up inventory of tritium are injected into the plasma. During the burn stage, fusion reactions release 17.6 MeV of energy, carried primarily by high-energy neutrons. These neutrons drive the breeding stage, where they interact with lithium-6 in the surrounding blanket to create new tritium and helium. Finally, in the extraction and recycling stage, tritium is separated, purified, and reinjected to sustain the plasma.

⁶³ FAS, n.d.

⁶⁴ NRC, n.d.

⁶⁵ FAS, n.d.

To maintain long-term operation, a tritium breeding ratio (TBR) of at least 1.1 is required—ensuring sufficient fuel to cover decay, processing losses, and operational reserves (IAEA, n.d.).

Why It Matters for Commercialization

Once breeding systems are optimized, fuel costs become minimal, deuterium is inexpensive, and lithium-derived tritium adds only a small premium per megawatt-hour. However, breeding blankets and associated systems can represent 10–15% of first-of-a-kind (FOAK) capital expenditure (CAPEX). Companies that achieve high-throughput tritium extraction, develop advanced permeation barriers, or deploy innovative LiPb pumping systems will be well-positioned to secure early licenses and reduce plant costs.⁶⁶

Risk Framing: The Cost of Breeding Underperformance

If breeding blankets fail to achieve a tritium breeding ratio of at least 1.1, plants will be forced to rely on external tritium purchases. With global civilian-accessible inventories below 25 kilograms today, this path is not scalable. Breeding inefficiency would also increase the levelized cost of electricity (LCOE) by driving dependence on scarce tritium or costly enrichment programs, eroding fusion’s competitive advantage over renewables and advanced fission. Regulatory barriers add further risk, as nuclear agencies such as the NRC and IAEA will require demonstrated control of tritium permeation and emissions before granting approvals. Finally, without clear evidence of a closed fuel cycle, investors and lenders are unlikely to finance first-of-a-kind plants. Together, these risks underscore that breeding performance is not a secondary challenge but a make-or-break determinant of fusion’s commercial trajectory.

Upside Framing: The Prize of Breeding Success

Conversely, if breeding blankets operate reliably, the prize is transformative. Fusion fuel supply would become effectively limitless, drawing on inexpensive deuterium from seawater and abundant global lithium reserves. With deuterium costs negligible and lithium-derived tritium adding only a marginal premium, fuel economics would fade as a barrier, shifting attention to optimizing capital and operating expenditures. Proven breeding and containment systems would also strengthen regulatory confidence, streamline licensing approvals, and improve public acceptance of fusion as a safe energy source. Just as importantly, demonstrating tritium self-sufficiency would de-risk the technology for institutional capital, unlocking financing for both first-of-a-kind and scaled deployments. In short, solving tritium breeding turns fusion from a physics experiment into a bankable global energy industry.

⁶⁶ FAS, n.d.

3.2 Critical Materials & Supply-Chain Constraints

The scalability of commercial fusion will hinge not only on physics demonstrations but also on the availability, cost, and manufacturability of several high-value, low-volume materials and components. Four categories stand out as the most immediate bottlenecks for first-of-a-kind (FOAK) and early fleet plants: high-temperature superconductor (HTS) tape, radiation-resistant structural alloys, precision-assembly capacity, and high-power laser diodes.⁶⁷

High-Temperature Superconductors (HTS). REBCO (rare-earth barium copper oxide) tape enables the high magnetic fields needed for compact-tokamak designs. Yet today's global output—just a few million meters annually—is insufficient to support more than one major compact-tokamak project at a time (ARPA-E). Industry forecasts suggest cumulative demand will exceed 200 million meters by the early 2030s, requiring a >20× scale-up. At present, costs of about US \$300 per kA·m, REBCO contributes heavily to FOAK plant capital costs. Business models assume declines to US \$10–20 per kA·m, but realizing that drop demands multi-billion-dollar investments in new production lines, none of which are yet under construction.

Structural Alloys. For first-wall and blanket structures, EUROFER97—a reduced-activation ferritic–martensitic steel—is the leading candidate, experimentally qualified to around 15 displacements per atom (dpa) before embrittlement and weld degradation become limiting.⁶⁸ Alternative alloys such as China's CHSN01 have demonstrated exceptional magnetic and mechanical strength (withstanding up to 20 tesla and 1.5 GPa), but their neutron irradiation performance under fusion fluences remains unproven.⁶⁹ Scaling either material to the thousands of tonnes required per plant would strain global specialty-steel melt and forge capacity.

Precision Manufacturing. Fusion components demand unprecedented assembly accuracy. Blanket and divertor modules, weighing tens of tonnes, must be positioned inside the vacuum vessel with tolerances tighter than 0.5 mm to align coolant channels, diagnostic ports, and tritium-handling systems. European demonstration programs have employed robotic movers and satellite-grade metrology to achieve this level of precision.⁷⁰ Delays in expanding high-spec machining and tooling capacity would directly translate into project slippage.

High-Power Laser Diodes. For laser-driven inertial confinement fusion (ICF), high-power diode arrays are the most constrained input. Current pulsed stacks operate at only ~5–7% wall-plug efficiency, while economic designs require 20–30%.⁷¹ Global supply is thin, with only a handful

⁶⁷ MIT, 2024; CATF, 2025

⁶⁸ MIT, 2024

⁶⁹ CATF, 2025

⁷⁰ MIT, 2024

⁷¹ CATF, 2025

of manufacturers capable of multi-megawatt arrays. Fusion-related diode demand remains under US \$20 million annually, far below the volume required to drive cost declines. Significant new investment in epitaxy and automated packaging facilities will be required to prevent this supply chain from becoming a gating factor for ICF commercialization.

Aggregate Risk. Collectively, these constraints form a critical-path risk for early deployment. Without synchronized investment in HTS tape production, radiation-hard alloy manufacturing, precision assembly infrastructure, and diode fabrication, even technically successful pilot plants could face multi-year delays or cost overruns that erode investor confidence. Proactively addressing these bottlenecks is essential if the industry is to meet commercialization targets in the 2030s.

3.3 Balance-of-Plant & Cogeneration Opportunities

Once plasma ignition is achieved, the downstream systems of a fusion plant closely resemble those of advanced thermal power facilities. Fusion's distinguishing feature is its high operating temperature — typically 500–800 °C — which allows integration with next-generation power cycles and multiple industrial cogeneration pathways. This enables developers to diversify revenues, enhance internal rates of return, and improve project economics.

Supercritical CO₂ Power Cycles. At ~25 MPa and ~550 °C, sCO₂ Brayton cycles can reach 45–50% thermal-to-electric conversion efficiency, while occupying only one-quarter of the footprint of conventional steam systems. Because they eliminate water makeup, they are especially attractive for arid or inland sites. Vendors including GE Vernova and Hanwha are testing 10–50 MW prototypes, with commercial-grade units expected to be available by the time first fusion pilots reach operation.

High-Temperature Heat Exchangers. Efficient transfer of hundreds of megawatts from coolants such as FLiBe, Li-Pb, helium, or high-pressure CO₂ into secondary cycles will rely on printed-circuit and diffusion-bonded heat exchangers made from nickel-based superalloys (e.g., Haynes 230, Alloy 800H). These designs can withstand >700 °C pressurized flows while minimizing footprint. Scaling them to hundreds of tonnes per plant will require expanded forging and hot isostatic pressing capacity, benefitting both fusion and adjacent industries like advanced fission and hydrogen reforming.

Hydrogen and Ammonia. Outlet temperatures above 650 °C align with solid-oxide electrolysis cells (SOECs), which reduce electricity demand by ~20–30% compared to PEM systems. A 1 GWe fusion plant could produce on the order of 180–220 kt of hydrogen per year, sufficient to yield about 1 Mt of ammonia via Haber–Bosch. These volumes are material for fertilizer markets and as a zero-carbon fuel for maritime transport.

Desalination and District Energy. Low-grade heat at 80–120 °C, recovered from sCO₂ reject stages, can drive multi-effect or membrane distillation systems, producing 2–6 million m³/day of

freshwater from a 1 GWe plant without reducing net output. Inland facilities could instead use this heat for district cooling via absorption chillers, boosting utilization and reducing natural gas reliance.

Direct Air Capture (DAC). Fusion's steady power and high-temperature output also support advanced DAC chemistries such as calcium looping, which require 600–900 °C regeneration heat. Co-located DAC could remove up to ~0.3–0.5 t CO₂ per MWh(th), generating premium carbon credits and diversifying revenues.

Integration Impact. Well-structured cogeneration strategies could improve project IRRs by several percentage points: 3–5 points from hydrogen or ammonia contracts, 1–2 points from water or district energy in resource-constrained regions, and incremental upside from carbon-credit sales. Because these subsystems are already proven in adjacent industries, they present relatively low technology risk and can be modularly deployed. This positions fusion not just as an electricity supplier but as a multi-commodity energy hub, enhancing resilience and broadening addressable markets.

3.4 Commercialization Roadmaps

Fusion commercialization is advancing along parallel tracks: national programs, multinational initiatives, and ambitious private-sector efforts. Collectively, these roadmaps aim to bridge the gap between experimental proof-of-concept devices and grid-connected, revenue-generating plants within the next two decades.

United States – DOE Milestone Program. The Department of Energy's *Milestone-Based Fusion Development Program* authorizes up to \$415 million through FY2027 across eight private developers. The program funds staged progress toward preliminary plant designs and licensing roadmaps, with future phases contingent on congressional appropriations. An initial \$46 million tranche was awarded in mid-2024 to support Phase 1 activities.

United Kingdom – STEP Prototype. The UK government has committed £2.5 billion to the *Spherical Tokamak for Energy Production (STEP)* project, to be built at the former West Burton A coal station in Nottinghamshire. Current plans target groundbreaking in 2028, first plasma in 2035, and initial electricity generation in the early 2040s.

European Union – DEMO Program. EUROfusion's roadmap is anchored by ITER, now targeting first plasma in 2035. Key enabling experiments include IFMIF-DONES in Spain, which began construction in 2023 and is scheduled for operation in the early 2030s to qualify structural materials. Based on ITER and IFMIF data, a 300–500 MWe DEMO plant could enter licensing in the early 2040s, with routine generation before mid-century.

Private-Sector Timelines.

- Commonwealth Fusion Systems (HTS tokamak): *SPARC* first plasma projected 2026–27; *ARC* demonstration in the early 2030s.
- Helion Energy (pulsed FRC): *Microsoft PPA* for 50 MW, with deliveries targeted in 2028; site preparation began in Everett, WA, in mid-2025.
- General Fusion (magnetized target): 70% scale non-electric demonstrator at Culham, UK, commissioning 2026–27. No published CAPEX or LCOE targets.
- Zap Energy (sheared-flow Z-pinch): *FuZE-Q* aims for $Q \approx 1$; commercial power rating not yet public.
- Xcimer Energy (laser ICF): *Vulcan* multi-megajoule laser expected by ~2030; grid power targeted for the mid-2030s.

Pilot Plant Siting Realities. All first-wave commercial plants face several gating requirements: securing start-up tritium inventories (kg-scale), obtaining 300–400 kV grid interconnections that require long-lead permitting, and meeting NRC Part 30 or equivalent national limits on tritium release and activated material handling. For these reasons, developers favor brownfield industrial sites such as retired coal plants or national laboratory campuses, which already offer grid connections, water rights, transportation access, and experienced workforces. These siting advantages shorten project timelines and reduce regulatory risk.

3.5 Economics & Financing Structures

The transition from pilot-scale fusion facilities to grid-connected commercial plants will depend on meeting cost targets, securing bankable offtake agreements, and structuring finance to handle high capital intensity and long build timelines.

Capital Requirements by Technology Pathway. Estimated first-of-a-kind (FOAK) CAPEX varies significantly by approach. High-field tokamaks such as Commonwealth Fusion Systems' *ARC* design project cost \$2–3 billion per GW-electric (GWe), driven by HTS tape and breeding blanket systems. Magnetized target fusion concepts (e.g., General Fusion) aim for \$1.5–1.7B/GWe, avoiding superconducting magnets. Pulsed field-reversed configuration (FRC) systems, such as Helion's target ~\$1B/GWe, leverage direct electricity conversion instead of steam turbines. Laser inertial confinement fusion (ICF), such as Xcimer Energy, remains at the upper end, \$3–4B/GWe, reflecting diode and target factory costs. Compact Z-pinch systems (Zap Energy) are the most speculative, with potential for sub-\$1B/GWe if electrode lifetimes can be validated. These figures should be seen as indicative, as none have yet been demonstrated at the plant scale.

Learning Curves and Cost Reduction. Developers commonly apply learning rates of 14–17% per cumulative capacity doubling, drawing from offshore wind and advanced gas turbines. If achieved, three doublings (~10–15 plants) could cut high-field tokamak costs below \$1.7B/GWe and pulsed-magnet systems below \$700M/GWe. At such levels, fusion would compete with

small modular reactors (SMRs) and natural gas with CCS, though it remains unproven whether large nuclear-class projects can replicate the same learning dynamics as modular renewables.

Levelized Cost of Electricity (LCOE). Most pathways target \$40–60/MWh for tokamak and pulsed-magnet designs, and \$55–75/MWh for magnetized target and laser systems once fleet learning takes hold. These assume 30-year plant life, 90% capacity factor, and 7–8% WACC. Policy incentives such as the U.S. Inflation Reduction Act’s \$45Y/\$48E credits could lower LCOE by 10–15%, while the UK’s Contracts for Difference (CfD) program provides price certainty to improve bankability.

Emerging Revenue Models. The Helion–Microsoft 50 MW PPA (2023) marked the first long-term fusion power contract. Other utilities, including ENI (CFS investor) and Duke Energy (Helion MOU), are exploring option-based PPAs triggered by pilot milestones. These structures blend venture-style upside with infrastructure-grade downside protection, creating revenue visibility years before operations and supporting project finance.

Financing Structures. Until fusion demonstrates multi-month sustained operations above $Q=1$, project financing remains equity-heavy, sourced from venture capital, corporate strategics, and grants (e.g., DOE Milestone Program, UK STEP). Transitioning to limited-recourse debt will require: (1) proven sustained $Q > 1$ operations, (2) long-term offtake contracts with investment-grade counterparties, and (3) credit-wrapped EPC and insurance packages. At that point, fusion projects could support 50–60% leverage, initially via green bonds or export credit agencies, maturing into syndicated infrastructure loans once operational data de-risks the asset class.

Policy-Driven Cost Advantages. In the U.S., \$45Y and \$48E incentives can cut early-fleet LCOEs by 10–15%, while the UK’s CfD framework underwrites price certainty to support debt. Coupled with supply-chain localization, these policy levers could accelerate FOAK cost-down trajectories.

Strategic Considerations for Investors. Fusion’s economic case rests on three pillars: hitting early CAPEX targets, locking in long-term offtake agreements, and achieving predictable cost-down curves. By combining tax incentives with anchor customers in premium early markets, developers can front-load returns, bolster investor confidence, and accelerate the shift from equity-led to debt-financed projects.

4. Market Opportunities & Value Creation

4.1 Early-Market Applications and End-Use Cases

Fusion’s capacity to deliver constant, high-grade heat and electricity at scale opens a broad spectrum of commercial applications that extend beyond grid power. Many of these use cases are highlighted in existing industry roadmaps and early market engagement from developers and corporate off takers. Because the sector’s technological diversity includes steady-state

tokamaks, pulsed-magnet systems, magnetized-target fusion, laser inertial, and compact Z-pinch devices, the application range is broader than for most other clean-energy technologies.⁷²

Grid Electricity Supply

Fusion reactors designed for baseload operation can achieve capacity factors above 90%, supplying continuous zero-carbon electricity to regional and national grids. The Microsoft–Helion 50 MW power purchase agreement (PPA), targeting first delivery in 2028, is the sector’s first long-tenor commercial contract, demonstrating that large, creditworthy buyers will commit to fusion power well before commissioning if milestones are explicit.⁷³ Similar agreements under discussion with utilities and heavy industry would further validate bankable revenue models.

Green Hydrogen and E-Ammonia Production

Outlet temperatures in the 600–700 °C range enable direct coupling with solid-oxide electrolysis cells (SOECs), which require less electricity input than PEM systems. A single 1 GW-thermal fusion plant co-sited with SOEC units could yield about 200,000 t of hydrogen annually, enough to manufacture roughly one million tonnes of green ammonia for fertilizer or maritime fuel markets.⁷⁴ This integration can lower production costs and improve plant load factors by monetizing every megawatt-hour of thermal output.

Industrial Process Heat

Many industrial processes, including green-steel direct-reduced iron, cement production, and high-temperature chemical synthesis, require heat in the 500–700 °C range. Fusion plants can supply this heat directly or via heat exchangers, displacing fossil-fired boilers and enabling deep industrial decarbonization.⁷⁵ Lower-temperature loops (80–150 °C) rejected from primary power cycles can be cascaded into district heating or absorption chilling for agro-industrial complexes, food processing, and cold-storage operations.

Cogeneration and Multi-Commodity Hubs

Cogeneration designs leverage high-temperature sCO₂ Brayton cycles to produce both electricity and thermal energy for hydrogen, ammonia, desalination, or direct-air-capture (DAC) systems. Coastal fusion plants can drive multi-effect distillation or membrane distillation units to produce 2–

⁷² Fusion Industry Association, 2024 global fusion industry report.

⁷³ Helion Energy, “Helion announces world’s first fusion PPA.”

⁷⁴ MIT Energy Initiative, *The Role of Fusion*; Clean Air Task Force, *Fusion on the grid*.

⁷⁵ Clean Air Task Force, *Fusion on the grid*.

6 million m³/day of fresh water without significant electrical output loss, while inland projects can use waste heat for DAC processes that yield premium carbon credits.⁷⁶

Coal-Plant Repowering

The modularity of some reactor designs allows them to fit within the footprint of retired coal plants, re-using high-voltage interconnects, cooling water access, and transmission infrastructure. This approach shortens permitting timelines, maintains property-tax revenue for host communities, and keeps grid inertia in place, aligning with early site-selection strategies in the U.K. STEP programmed and U.S. coal-to-fusion conversions.⁷⁷

Maritime and Remote Power

Compact designs such as Z-pinch and Orbitron™ micro-fusion reactors, targeted at 5–20 MW-electric output, aim to fit within ISO containers. This format could replace bunker fuel on deep-sea vessels or diesel generation in remote mining, military, or disaster-relief operations.⁷⁸ Early versions can serve as transportable neutron sources for isotope production and non-destructive testing, providing interim revenue streams before propulsion-class certification.

Value Creation Mechanisms:

- Multi-product revenue streams increase overall plant profitability and resilience against electricity price fluctuations.
- Load-factor optimization from integrated industrial or hydrogen production raises capacity factors and revenue per installed MW.
- Reduced siting risk by targeting brownfield and maritime applications, which often have streamlined permitting.
- Carbon market participation through DAC or high-integrity carbon credits linked to zero-carbon industrial production.

In summary, the diversity of operational profiles and heat grades across fusion designs allows developers to tap into multiple high-value markets simultaneously. These applications broaden addressable revenue, improve return on investment, and accelerate the sector's transition from demonstration to bankable infrastructure.

Fusion's inherent attributes — near-continuous operation, high thermal output, compact footprint, and zero direct emissions — position it for specialized early markets willing to absorb first-generation power prices in the \$60–80/MWh range. Early deployments into these niches

⁷⁶ MIT Energy Initiative, *The Role of Fusion*.

⁷⁷ UK Government, "£2.5 billion for world-first prototype fusion energy plant," *GOV.UK*, June 12, 2025; Fusion Industry Association, *2024 global fusion industry report*.

⁷⁸ Fusion Industry Association, *2024 global fusion industry report*.

can create bankable revenue streams, accelerate technology maturation, and drive cost reductions through scaled production.

Always-On Data Center Power. Hyperscale cloud and AI campuses require firm, zero-carbon electricity to meet 24/7 carbon-free energy (CFE) commitments. Fusion’s high capacity factor (>90%) and modular scale (50–400 MWe) map well to hyperscale load profiles. The Helion–Microsoft 50 MW PPA, signed in 2023 for delivery in 2028, illustrates that blue-chip offtakers are willing to secure fusion supply years in advance — a key factor for project finance bankability.

Green Hydrogen and Ammonia Hubs. Fusion outlet temperatures of 500–700 °C enable integration with solid-oxide electrolysis cells (SOECs), which require ~20–30% less electricity than PEM electrolysis. A 1 GWe fusion plant could produce roughly 200,000 t of hydrogen per year, enough to synthesize ~1 Mt of green ammonia via Haber–Bosch. This configuration captures value from both electricity and heat, aligning with long-term offtake opportunities in fertilizer and maritime fuels.

Agro-Industrial Process Heat. Agriculture and food processing demand steady steam in the 150–350 °C range for ethanol distillation, pasteurization, and fertilizer production. Fusion’s low-grade reject heat from sCO₂ turbines can displace fossil boilers while also driving absorption chillers for cold storage. This creates integrated energy–refrigeration hubs for agricultural cooperatives.

Coal-Plant Repowering. Thousands of retired coal plants worldwide retain valuable grid ties, cooling water systems, and trained local labor. Fusion modules, requiring <10 hectares, could be sited on these brownfields, leveraging existing infrastructure, preserving tax bases, and avoiding protracted greenfield permitting.

Maritime and Remote Micro-Fusion. Compact designs under development range from kWe to tens of MWe. While large cargo ships require >50 MWe propulsion, early micro-fusion concepts are more likely to serve smaller vessels, auxiliary loads, offshore facilities, or remote industrial operations where diesel logistics are costly. Neutron-source variants could provide nearer-term revenue streams in medical isotope production and non-destructive testing. The IMO’s 2023 GHG Strategy, which targets net-zero “by or around 2050” with interim milestones, creates a long-term pull for such maritime innovations.

Commercial Takeaways. In each of these early markets, fusion’s reliability and modularity are valued as much as its emissions profile. Targeting customers with urgent decarbonization needs and high willingness-to-pay — from tech firms to fertilizer producers — enables developers to secure early PPAs and heat-offtake agreements. These contracts de-risk deployment, catalyze investor confidence, and accelerate fusion’s march from pilot projects to commercial readiness.

The market for fusion spans far beyond wholesale electricity. Because fusion aims to provide clean, firm power with high-grade heat and a compact site footprint, it unlocks premium early markets

aligned with decarbonization, energy security, and industrial competitiveness—while grid sales remain the anchor revenue stream (see also corporate 24/7 CFE procurement trends). [wri.org](https://www.wri.org)

24/7 clean power for hyperscale data centers. AI and cloud growth are pushing hyperscale loads into the tens to hundreds of megawatts, with many sites ≥ 100 MW—creating demand for round-the-clock, zero-carbon supply. Fusion’s firm output can match this profile without the integration complexity of pairing intermittent renewables with very large storage. The Helion–Microsoft 50 MW power purchase agreement (PPA), announced in 2023 for targeted delivery beginning 2028 (after a one-year ramp), demonstrates that major offtakers will contract for fusion capacity well ahead of commercial operation—supporting bankability. IEA Helion+1

Green hydrogen and ammonia. With advanced blankets/cycles, fusion plants could deliver high-temperature heat (~ 500 – 700 °C) and electricity suitable for solid-oxide electrolysis (SOEC), which typically operates at ~ 600 – 850 °C and can cut electrical consumption vs PEM by ~ 20 – 30% , lowering levelized hydrogen costs. At ~ 40 – 45 kWh/kg electricity input, ~ 1 GWe continuous output could produce ~ 190 – 220 kt H_2 /yr; that hydrogen could synthesize ~ 1.0 – 1.2 Mt NH_3 /yr via Haber–Bosch (~ 177 kg H_2 per tonne NH_3). These volumes are directly relevant to fertilizer supply chains and emerging zero-carbon shipping fuels. Catf+1 ISPTES Conferences

Agro-industrial heat cascades. Waste heat from power cycles (e.g., sCO_2) can be cascaded to low-temperature uses such as ethanol distillation, food sterilization, and absorption-chiller refrigeration—displacing gas boilers, lowering OPEX, and boosting overall plant utilization (integration depends on site-specific temperatures and layouts).

Coal-site repowering (brownfield siting). Thousands of retiring coal sites offer grid interconnections, cooling water, transport access, and skilled labor. Evidence from advanced nuclear SMR studies shows repowering can reuse infrastructure and shorten timelines; fusion developers could potentially pursue similar brownfield strategies as designs mature, with permitting advantages and preserved local benefits. [PNNL GAIN Nuclear Bipartisan Policy Center](https://www.pnnl.gov/publications/pnnl-gain-nuclear-bipartisan-policy-center)

Maritime and remote microgrids (select niches). Compact fusion concepts are diverging in scale: Zap Energy publicly targets ~ 50 MWe per core, suggesting potential for modular, port-adjacent power or smaller industrial loads, while Avalanche Energy’s “Orbitron” is 1–100 kWe, pointing to portable micro-fusion R&D. For large blue-water cargo ships, propulsion typically requires tens of MW to >80 MW, so early fusion applications are more plausible for auxiliary loads, smaller vessels, offshore platforms, or remote industrial microgrids until higher-output, marine-qualified reactors exist. Meanwhile, the IMO’s 2023 GHG Strategy sets net-zero “by or around 2050” with interim 2030/2040 ambitions—creating a pull for zero-carbon fuels (H_2 , NH_3) and clean firm power at ports. [Zap Energy](https://www.zapenergy.com) [Avalanche Energy](https://www.avalancheenergy.com) [International Maritime Organization](https://www.maritime-organization.org)

Bottom line. These segments create multiple independent demand streams capable of absorbing early fusion output at above-grid prices. By focusing on customers with urgent decarbonization needs, high willingness-to-pay, and clearer siting pathways, developers can accelerate learning curves, strengthen investor confidence, and compress time-to-scale—while using contractual precedents (e.g., fusion PPAs) to improve financeability. [Helion](#)

4.2 Investment Opportunities

Capital can target cross-cutting bottlenecks and enable near-term commercial steps across multiple reactor pathways. The following opportunities align with the citations already embedded in the paper and the sources they reference.

1) High-temperature-superconductor (HTS) manufacturing scale-up

Compact tokamaks depend on rare-earth barium copper oxide (REBCO) tape for high-field magnets. CFS’s 20 tesla magnet result and multi-billion-dollar roadmap signal bankable demand for large tape volumes, yet present industry pricing and throughput remain constraints.⁷⁹ ARPA-E and technical literature identify the cost target at roughly 10 dollars per kA-m to reach competitive plant economics, well below recent commercial ranges reported for REBCO tape.⁸⁰ Equity for new deposition lines, quality control, and long-length yields can earn scarcity premiums because each percentage cut in tape price translates into large reductions in tokamak FOAK capex and levelized costs.⁸¹

2) Tritium breeding, processing, and accountancy systems

Civilian tritium inventories are small and decay each year, which makes closed-loop breeding a prerequisite for commercial roll-out. Recent analyses estimate civilian stocks near 25 kilograms as of 2024, while plant-level needs are measured in kilograms per year per unit once at scale.⁸² Investment is needed in lithium-six enrichment, high-throughput extraction from liquid-metal or ceramic blankets, permeation-barrier materials, and real-time metrology and safeguards. These subsystems are licensable under the NRC’s fusion framework and will account for a meaningful share of FOAK capex, creating a classic pick-and-shovels position in the value chain.⁸³

⁷⁹ Commonwealth Fusion Systems, “CFS closes US\$1.8 billion Series B”; MIT Energy Initiative, *The Role of Fusion*.

⁸⁰ Advanced Research Projects Agency-Energy (ARPA-E), “BETHE: Breakthroughs enabling thermonuclear-fusion energy.”

⁸¹ Commonwealth Fusion Systems, “CFS closes US\$1.8 billion Series B”; MIT Energy Initiative, *The Role of Fusion*; ARPA-E, “BETHE.”

⁸² Kleinman Center for Energy Policy. (2025). *Energy policy reports*. <https://www.kleinmanenergy.upenn.edu>; Nuclear Engineering International. (2025). *Global nuclear fusion updates*. <https://www.neimagazine.com>; Federation of American Scientists. (2024). *Fusion energy development status*. <https://fas.org>

⁸³ U.S. Nuclear Regulatory Commission, Options for licensing and regulating fusion energy systems; Clean Air Task Force, Fusion on the grid.

3) High-power laser components and targets for inertial fusion

Laser-driven concepts require large improvements in wall-plug efficiency and drastic cuts in target cost to meet electricity price targets discussed in sector studies. LLNL ignition established scientific feasibility, and private efforts are pursuing long-pulse excimer or next-generation diode-pumped drivers to lift efficiency into the 20 to 30 percent band, which sector analyses identify as necessary for sub-75 dollars per MWh plants.⁸⁴ Investments in epitaxial throughput, bar packaging automation, thermal management, and high-rate target fabrication are leveraged across multiple inertial approaches.⁸⁵

4) AI-enabled plasma control, synthetic diagnostics, and digital twins

Software that shortens commissioning, expands safe operating windows, and reduces unplanned downtime scales with every additional reactor and requires modest capital relative to hardware. This segment benefits all major pathways, including HTS tokamaks, pulsed magnet field reversed configurations, magnetized target, and Z pinch. The policy community underscores the role of control and diagnostics in achieving earlier bankability, which ties directly to tax-credit timelines and offtake execution.⁸⁶

5) Early project-development equity anchored by long-tenor offtake

The first long-dated corporate offtake in the sector is the 50 MW Helion–Microsoft PPA, with Constellation as power marketer and initial delivery targeted for 2028. Site preparation for Helion’s first plant in Washington State began in July 2025. These facts provide a template for term sheets that activate upon technical milestones and for blended finance that admits infrastructure debt after initial runs above $Q > 1$ and license issuance under the NRC’s framework.⁸⁷ Project-level equity can convert to carried interests once tax-equity and limited-recourse debt enter, similar to early wind and solar plays but at larger unit sizes.⁸⁸

6) Supplier consolidation and service platforms

FIA’s 2024 survey shows over 7.1 billion dollars of cumulative private funding and rising public support. As FOAK programs move into detailed engineering, supplier consolidation will likely occur around HTS conductors, cryogenics, blanket modules, high-temperature heat exchangers,

⁸⁴ MIT Energy Initiative, *The Role of Fusion*.

⁸⁵ MIT Energy Initiative, *The Role of Fusion*.

⁸⁶ Clean Air Task Force, *Fusion on the grid*; Resources for the Future, *Projected impacts*.

⁸⁷ Helion Energy, “Helion announces world’s first fusion PPA”; Reuters, 2025; U.S. Nuclear Regulatory Commission, *Options for licensing*; Fusion Industry Association, *2024 global fusion industry report*.

⁸⁸ Fusion Industry Association, *2024 global fusion industry report*; Resources for the Future, *Projected impacts*.

and sCO₂ turbomachinery. Roll-up strategies and long-term offtake agreements can lock in manufacturing slots and pass learning-curve gains to plant developers and off-takers.⁸⁹

7) Policy-linked opportunities

U.S. tax credits under sections 45Y and 48E apply to fusion, which reduces levelized costs and improves early project returns according to energy-policy assessments. At the same time, investors should monitor proposals that would weaken or repeal these provisions, because modeled outcomes indicate higher retail electricity prices and weaker project economics if credits are withdrawn.⁹⁰ The NRC’s decision to license fusion under 10 CFR Part 30 lowers licensing risk and time compared with fission rules, which directly supports underwriting assumptions for FOAK projects.⁹¹

Why are these opportunities timely?

- Documented investment momentum and public funding have reduced early technical and policy uncertainty, which improves the odds that supply-chain investments will find near-term buyers among leading developers.⁹²
- HTS, tritium systems, laser photonics, and controls are pathway-agnostic levers that benefit several contenders and therefore diversify technical risk within a fusion-exposed portfolio.⁹³
- The first corporate PPA provides a reference structure for offtake and project SPVs, which is a necessary precursor to limited-recourse debt and tax-equity participation in the U.S. market.⁹⁴

Investment Opportunities and Key Value Drivers

The fusion sector offers diverse avenues for strategic investment across enabling materials, fuel-cycle infrastructure, subsystems, software, and early project finance. These opportunities span major reactor pathways — tokamak, magnetized target, pulsed FRC, and laser inertial — and provide investors with diversified exposure to the sector’s upside.

High-Temperature Superconductors. REBCO tape is the essential material for compact tokamaks, enabling the high magnetic fields that shrink reactor footprints. Current production, measured in only a few million meters annually, would support no more than a single ARC-class plant. Forecast demand exceeds 200 million meters by the early 2030s, requiring >20× scale-up.

⁸⁹ Fusion Industry Association, *2024 global fusion industry report*; European Commission, “ITER funding factsheet.”

⁹⁰ RFF, 2025

⁹¹ U.S. Nuclear Regulatory Commission, *Options for licensing*.

⁹³ MIT Energy Initiative, *The Role of Fusion*; Clean Air Task Force, *Fusion on the grid*.

⁹⁴ Helion Energy, “Helion announces world’s first fusion PPA”; Reuters, 2025

At today's ~\$300/kA·m, REBCO contributes heavily to plant CAPEX; each 10% price cut can save roughly \$100M per gigawatt-scale plant. Long-term supply deals, such as CFS's offtake agreement with Fujikura, suggest these inputs are already financeable.

Tritium Breeding and Handling. A 1 GWe fusion plant requires ~120 kg of tritium annually — five times the current global civilian inventory. Startups advancing lithium-6 enrichment, liquid-metal extraction modules, and low-permeation piping sit in a classic “picks-and-shovels” position. Even a 5% share of blanket-system spending (~\$150M per plant) could generate substantial returns as fleets scale. Policy clarity on civilian lithium-6 will be pivotal.

Laser-Diode Innovation. In ICF pathways, diode arrays account for ~⅓ of plant cost. Present wall-plug efficiencies (5–7%) must rise to 25–30% to hit sub-\$75/MWh economics. Advances in epitaxial GaAs growth, high-brightness pump designs, and automated packaging could simultaneously cut CAPEX and parasitic load. Scaling supply chains to multi-GW markets remains the parallel challenge.

AI-Driven Control and Digital Twins. Real-time AI plasma control, synthetic diagnostics, and digital twins can shorten commissioning cycles, extend component lifetimes, and maximize Q values. These platforms are capital-light, IP-intensive, and suited to SaaS business models that grow with deployments. Early traction under DOE's AI2ES program and partnerships with cloud providers shows commercialization pathways opening before full-scale plants arrive.

Project-Development Equity. The Helion–Microsoft 50 MW PPA (2023) marked the first fusion power contract, signaling a new asset class. Early equity stakes in project SPVs can convert into carried interests or preferred returns once limited-recourse debt is available, mirroring early wind/solar development — though with larger ticket sizes and higher technical barriers.

Summary. Five areas — HTS magnets, tritium infrastructure, diode efficiency, AI software, and project finance structures — represent critical levers for determining which fusion developers reach the grid first and at what cost. Strategic capital across these domains offers investors diversified exposure to fusion's aggregate success, regardless of which reactor design dominates.

4.3 Key Value Drivers

The commercialization trajectory for fusion energy will be shaped by several quantifiable factors that directly influence cost, deployment timelines, and investor confidence. These drivers cut across all major technological pathways, making them critical for both developers and investors evaluating the sector.⁹⁵

⁹⁵ CATF, 2025

1) Fiscal Incentives and Policy Support

The U.S. Inflation Reduction Act's technology-neutral \$45Y production and \$48E investment tax credits apply to fusion facilities commissioned after 2025. Depending on wage and domestic content compliance, these incentives can reduce the levelized cost of energy (LCOE) by an estimated 10–20% during the first decade of operation.⁹⁶ Treasury's 2025 guidance confirmed fusion's eligibility, providing long-term policy certainty that underpins bankable project finance structures.⁹⁷ Comparable mechanisms in the U.K., such as Contracts-for-Difference (CfD) arrangements, offer similar price stability for early plants.⁹⁸

2) Regulatory Clarity

The U.S. Nuclear Regulatory Commission's decision to license fusion under 10 CFR Part 30, a framework designed for byproduct materials, significantly shortens permitting timelines compared to fission rules. This shift reduces compliance costs and accelerates time-to-market for first-of-a-kind (FOAK) projects.⁹⁹ The ADVANCE Act of 2024 further reinforced this approach by requiring the NRC to maintain a distinct regulatory lane for fusion and to report regularly on licensing progress.¹⁰⁰ In practice, regulatory certainty de-risks project schedules, a key factor for both equity and debt investors.

3) Corporate Demand for 24/7 Clean Power

Data centers, hydrogen hubs, and energy-intensive industries require high-reliability zero-carbon energy. Long-term power purchase agreements (PPAs) from these sectors establish bankable revenue floors before plant commissioning. The 50 MW Helion–Microsoft PPA, scheduled for initial delivery in 2028, provides the first proven structure for long-tenor offtake in fusion.¹⁰¹ Similar term sheets are reportedly being explored by utilities and heavy industry, signaling the potential for broader market adoption of this model.

4) Capital Momentum

Large equity rounds validate developer credibility and attract additional funding. Commonwealth Fusion Systems' US \$1.8 billion Series B and Helion's multiple large raises are prime examples of how capital concentration accelerates both core technology and supply-chain readiness.¹⁰² This capital momentum also influences supplier investment decisions,

⁹⁶ RFF, 2025

⁹⁷ RFF, 2025

⁹⁸ UKG, 2025

⁹⁹ NRC, 2023

¹⁰⁰ U.S. Nuclear Regulatory Commission. (2024, February). Options for licensing and regulating fusion energy systems (10 CFR Part 30 pathway). <https://www.nrc.gov>

¹⁰¹ Helion Energy, 2023

¹⁰² CFS, 2021; FIA, 2024

particularly for critical components such as high-temperature superconductors (HTS) and tritium systems.

5) System-Wide Abatement Economics

According to the MIT Energy Initiative, the availability of commercial fusion could reduce global decarbonization costs by US \$3.6–8.7 trillion over the century. This macroeconomic impact strengthens fusion’s position in policy discussions and carbon-market strategies. Ultra-low lifecycle emissions ($< 5 \text{ g CO}_2\text{-e/kWh}$) make fusion competitive under stringent emissions standards, while its high-capacity factor ($> 90\%$) reduces the need for expensive long-duration storage.¹⁰³¹⁰⁴

6) Supply Chain and Component Readiness

Investments in enabling technologies, such as HTS magnets, tritium breeding blankets, supercritical CO₂ turbines, and AI plasma-control systems, are pathway-agnostic and benefit multiple reactor designs. Early supplier commitments and long-term offtake agreements can help avoid bottlenecks that would otherwise delay deployment.¹⁰⁵

A. Headwinds

Persistent technical risk, demonstrating sustained $Q > 1$ operation, qualifying neutron-resistant materials, and integrating tritium breeding, remains the principal hurdle. Supply constraints for HTS tape and specialized cryogenics threaten to delay scale-up if capacity is not added in the next five years. FOAK capital expenditure is still projected at US \$2–4 billion per gigawatt, making projects sensitive to interest-rate swings. Competing firm-power options (advanced fission, enhanced geothermal) are also racing to capture the same market niches. Finally, while fusion’s radiological profile is benign compared with fission, unfamiliarity can trigger local siting opposition.

B. Tailwinds

Several structural, technological, and policy factors are accelerating the trajectory toward commercial fusion deployment. These tailwinds reduce early-stage risk, improve investor confidence, and increase the probability of first-of-a-kind (FOAK) plants reaching operational status in the 2030s.

1) Strong Governmental Support

¹⁰³ MIT, 2024

¹⁰⁴ FIA, 2024

¹⁰⁵ FIA, 2024; MIT, 2024

National governments are committing significant funding to fusion development. In the United Kingdom, the government has allocated £2.5 billion to the STEP prototype at West Burton A, with the explicit aim of capturing domestic supply-chain benefits and achieving early demonstration of commercial viability.¹⁰⁶ In the European Union, ITER remains a cornerstone programmed with €5.6 billion committed for the 2021–2027 period, ensuring continuity of scientific and engineering research at the highest international level.¹⁰⁷ In the United States, the Department of Energy’s Milestone-Based Fusion Development Program is providing up to US \$415 million through FY 2027 to private developers, supporting pilot plant designs and hardware development.¹⁰⁸

2) Favorable Regulatory Frameworks

The U.S. Nuclear Regulatory Commission’s adoption of the 10 CFR Part 30 licensing pathway for fusion, distinct from fission’s more stringent rules, has significantly reduced licensing complexity and cost.¹⁰⁹ The ADVANCE Act of 2024 codified this approach, making it unlikely to be reversed in the near term.¹¹⁰ This framework shortens permitting timelines to an estimated 3–4 years, a substantial advantage in capital-intensive infrastructure projects.

3) Technological Breakthroughs in Enabling Components

Recent advances in high-temperature superconducting (HTS) magnets, as demonstrated by Commonwealth Fusion Systems’ 20-tesla large-bore magnet, have enabled more compact and potentially lower-cost tokamak designs.¹¹¹ In parallel, inertial confinement fusion has achieved scientific breakeven at Lawrence Livermore National Laboratory’s NIF, proving net-energy gain in a controlled experiment for the first time.¹¹² Such achievements enhance credibility across the sector and attract both public and private investment.

4) Corporate Offtake Commitments

The first long-dated corporate PPA in the sector, the 50 MW Helion–Microsoft agreement, has created a reference model for commercial contracting in fusion.¹¹³ This deal signals to the market that large, creditworthy buyers are willing to commit to fusion power years before plant commissioning, helping developers secure financing and lock in early revenue streams.

5) Rising Private Capital Inflows

¹⁰⁶ UKG, 2025

¹⁰⁷ EC, n.d.

¹⁰⁸ U.S. Department of Energy. (2024, January). Fusion energy development and milestones. <https://www.energy.gov>

¹⁰⁹ NRC, 2023

¹¹⁰ NRC, 2024

¹¹¹ CFS, 2021; MIT, 2024

¹¹² LLNL, 2022

¹¹³ Helion Energy, 2023

Private investment has reached US \$7.1 billion in cumulative funding as of 2024, with US \$900 million raised in the previous year alone.¹¹⁴ This capital influx is spread across diverse technologies, from large-scale tokamaks to micro-fusion systems, creating a robust innovation pipeline. The trend is reinforced by climate-focused mega-funds and strategic corporate investors who view fusion as a critical part of the long-term energy mix.

6) Growing Need for Firm, Zero-Carbon Power

As renewable penetration increases, grid operators face challenges in maintaining reliability without resorting to fossil backup. Fusion's ability to provide 24/7 zero-carbon baseload power directly addresses this need, making it an attractive complement to wind, solar, and storage in a balanced energy portfolio.¹¹⁵

Regulatory and Policy Shifts

United Kingdom: Developers can access the Contracts for Difference (CfD) scheme, which offers 15-year strike-price guarantees that are critical for securing infrastructure debt in FOAK projects.¹¹⁶

European Union

The European Union regulates fusion under the Basic Safety Standards Directive without technology-specific rules, leaving facility licensing to individual member states. Financially, the EU supports fusion through ITER contributions of €5.6 billion between 2021 and 2027 and Horizon Europe grants to EUROfusion. The forthcoming Net-Zero Industry Act could extend tax credits or green-bond guarantees to commercial fusion facilities in the late 2020s.¹¹⁷

Implications for Developers and Investors

In the near term, the U.S. offers the clearest path to bankable FOAK projects, combining a light regulatory framework with substantial tax incentives and a federal cost-share programmed. State-level enhancements in Washington, New Jersey, and potentially Massachusetts can further improve project economics. The U.K. offers revenue certainty via CfDs and strong industrial policy support, while the EU remains a major public funder of R&D but has yet to harmonize commercial licensing regimes. Investors should track:

1. Congressional budget appropriations for the DOE fusion programmed.
2. Any legislative changes to §45Y and §48E that could reduce credit values or transferability.

¹¹⁴ FIA, 2024

¹¹⁵ MIT, 2024; CATF, 2025

¹¹⁶ UKG, 2025

¹¹⁷ EC, n.d.

3. Emerging national or state incentives that target specific components such as HTS magnets or tritium systems.

Regulatory and Policy Shifts

The regulatory environment for fusion energy is evolving rapidly, with leading jurisdictions establishing licensing pathways and fiscal incentives to accelerate commercialization. The United States currently offers the most financeable and technology-specific framework, while the United Kingdom and European Union are developing parallel approaches with varying degrees of clarity and support.

United States. In 2024, the Nuclear Regulatory Commission (NRC) voted to regulate fusion power plants under 10 CFR Part 30, the “byproduct material” framework historically applied to industrial accelerators. This designation removes most prescriptive requirements designed for fission reactors, reducing licensing timelines from a decade to roughly three to four years and lowering compliance costs by an estimated 30 percent. The bipartisan ADVANCE Act of 2024 reinforced this approach, directing NRC to maintain a dedicated fusion pathway, create milestone-based licensing schedules, and expedite reviews of international designs.

The Inflation Reduction Act (IRA) extends tech-neutral clean energy credits to fusion: the \$45Y production tax credit provides up to 2.5¢/kWh for ten years, while the \$48E investment tax credit covers up to 30% of project CAPEX. Together, these incentives can reduce LCOE by 10–15%.

State-Level Incentives. Policy competition is emerging among U.S. states. Washington State expedited permitting for Helion Energy’s first plant and granted sales tax exemptions. New Jersey’s Fusion Technology Industry Commission Act (2023) offers R&D grants and payroll tax credits to attract suppliers. In Massachusetts, pending legislation would expand clean energy credits to cover HTS magnet and tritium-handling facilities, echoing its earlier support for offshore wind supply chains.

United Kingdom. The UK classified fusion separately from fission in 2022 guidance, with regulation overseen by the Health & Safety Executive. It has committed £2.5 billion to the STEP prototype at West Burton A. Fusion projects can access 100% capital allowances and revenue stability through the Contracts for Difference (CfD) mechanism, which guarantees strike prices for up to 15 years — a critical enabler for debt financing.

European Union. Within the EU, fusion facilities fall under the Euratom Basic Safety Standards Directive, leaving licensing to member states. Financial support is channeled primarily through ITER (€5.6B from 2021–2027) and Horizon Europe’s EUROfusion grants. The proposed Net-Zero Industry Act may extend green tax credits or guarantee mechanisms to fusion in the late 2020s, but as of 2025 no union-wide, technology-specific licensing framework has been codified.

Implications. The U.S. offers the most predictable environment for FOAK projects, combining a light-touch regulatory pathway with transferable tax credits. State-level incentives further reduce capital costs and timelines. The UK's CfD-backed model provides long-term price certainty, while the EU's emphasis on research funding underwrites technology maturation but leaves commercial licensing fragmented. Developers that secure early permits, tax elections, and offtake contracts in these lead markets will enjoy cost and schedule advantages difficult for later entrants to replicate.

5. Strategic Fit & Impact

5.1 Fit with Investment Portfolio

Nuclear fusion aligns closely with core investment theses around adaptation, mitigation, resilience, and the pursuit of outsized returns in mission-critical markets. Compared to most climate-technology verticals, fusion offers a uniquely comprehensive fit across all of these dimensions.

Fusion represents the essence of transformative investment: tackling humanity's most urgent challenges with bold, enduring solutions. It has the potential to deliver zero-carbon, virtually limitless baseload power—supporting planetary health while embedding systemic resilience into the global energy future. It is precisely the kind of assumption-breaking technology that demands ambitious founders, and it offers not just climate and energy benefits, but also the potential to create one of the most enduring and impactful businesses of our time.

On the mitigation side, a commercial fusion plant produces zero direct CO₂, methane, or particulate emissions. Life-cycle analyses place its carbon footprint at below 5 g CO₂-e/kWh, a level comparable to onshore wind yet delivered at far higher capacity factors exceeding 90 percent.¹¹⁸ This combination positions fusion as a foundational technology for deep decarbonization of both electricity grids and hard-to-abate sectors, including green hydrogen, synthetic fuels, and high-temperature industrial heat. On the resilience front, the near-unlimited supply of deuterium and on-site tritium breeding strengthens energy security by insulating communities and industries from fossil-fuel price volatility and geopolitical supply shocks.

From a portfolio construction perspective, fusion would expand exposure to firm, dispatchable clean power, providing a critical complement to existing positions in variable renewables, energy storage software, and grid flexibility tools. Fusion's output characteristics also unlock entirely new revenue streams—such as process heat for industrial applications, desalination, and 24/7 data-centre power—that remain inaccessible to wind, solar, or even advanced battery storage. Early

¹¹⁸ MIT, 2024

corporate offtake agreements, like Microsoft’s 50 MW PPA with Helion Energy, highlight a pathway to bankable, infrastructure-grade cash flows that align well with portfolio strategies emphasizing predictable, long-term revenue profiles.

The sector’s supply-chain gaps, notably in high-temperature superconductor (HTS) tape production, tritium-handling systems, supercritical CO₂ turbines, and AI-enabled plasma control, mirror the “critical component” strategy the firm has successfully applied in its power-electronics and hydrogen portfolios. Investing in these enabling technologies allows value capture across multiple reactor architectures, regardless of which ultimately dominates the market.

Momentum in the sector is amplified by favorable U.S. Nuclear Regulatory Commission licensing under 10 CFR Part 30, federal tax incentives under §45Y and §48E, and DOE milestone-based funding programmed. Together, these create a public-capital multiplier effect, lowering the risk profile for first-of-a-kind assets and improving the attractiveness of early-stage equity positions.

Long-term, if fusion developers meet cost targets below US \$50/MWh, global demand could surpass one terawatt of installed capacity by mid-century, representing an addressable market in the trillions of dollars. Entering the ecosystem at this stage offers an asymmetric return profile akin to early investments in lithium-ion batteries or utility-scale solar, but with greater market potential and deeper technology defensibility.

In short, strategic exposure to fusion and its enabling supply chains would round out a climate-focused portfolio, reinforce the resilience theme, and position investors at the forefront of the next generation of zero-carbon infrastructure.

5.2 Environmental and Climate Impact

Fusion energy offers a unique combination of ultra-low life-cycle greenhouse gas emissions, compact land requirements, system-wide decarbonization benefits, and resilience against climate shocks. These attributes position it as a critical contributor to global climate strategies while enhancing infrastructure reliability.

Ultra-Low Life-Cycle Carbon Intensity. Recent life-cycle assessments estimate that mature fusion plants will emit <5 g CO₂e/kWh, comparable to or slightly below advanced fission and leading renewables.¹¹⁹ Modeling for U.S. grids shows fusion remains competitive even under stringent ceilings of ≤20 g CO₂e/kWh, requiring minimal “carbon headroom” to qualify as zero-emission generation.

Minimal Land Footprint. Fusion reactors concentrate gigawatt-scale thermal output in buildings no larger than modern sports arenas. This yields a land-use intensity of <1 m² per MWh

¹¹⁹ MIT, 2024; CATF, 2025

generated annually, compared to $>10 \text{ m}^2/\text{MWh}$ for utility-scale solar.¹²⁰ For nations with limited land or biodiversity preservation goals, this compact footprint is a decisive advantage.

System-Wide Decarbonization Value. A 2024 MIT Energy Initiative study found that incorporating firm fusion plants into global mid-century portfolios could reduce total decarbonization costs by \$3.6–8.7 trillion. Fusion’s high capacity factor reduces reliance on long-duration storage, backup fossil plants, and extensive transmission buildouts, cutting overall system costs and increasing reliability.

Adaptation and Resilience Co-Benefits. Fusion plants are enclosed, hardened facilities that do not require continuous fuel deliveries, making them less vulnerable to disruptions from hurricanes, wildfires, or droughts that can curtail renewable output. Their $>500 \text{ }^\circ\text{C}$ high-temperature output can support district heating during winter storms and absorption cooling during heatwaves, sustaining community resilience. Co-located systems for desalination or direct-air capture further enhance value, enabling fusion complexes to function as integrated energy–water–carbon hubs in regions facing climate-driven stresses.

Summary. Fusion combines near-zero emissions, compact siting, multi-trillion-dollar system savings, and resilience benefits in a single scalable technology. Few other options offer this blend of deep decarbonization potential and high reliability, making fusion a cornerstone candidate for the clean energy transition.

5.3 Risk Landscape

The path to commercial fusion energy is shaped by a convergence of scientific, engineering, supply-chain, financial, and public-acceptance risks. Each must be managed to ensure timely deployment, cost competitiveness, and investor confidence.

Physics Validation. While the National Ignition Facility (NIF) has achieved $Q > 1$ in inertial confinement fusion under carefully tuned laboratory conditions, this milestone has not yet been replicated in a power-plant environment. Most private devices remain below commercial breakeven. The next milestone is to demonstrate sustained multi-hour operation, closed-loop tritium recycling, and resilience to high neutron fluence. Failure to achieve any of these would slow deployment trajectories.

Supply-Chain Bottlenecks. Fusion depends on inputs produced in highly concentrated markets. A small number of firms dominate global production of REBCO high-temperature superconducting tape, essential for compact tokamaks. Multi-megawatt laser diode arrays, critical for inertial fusion, are manufactured by only a handful of suppliers. Enriched lithium-6, necessary for tritium breeding,

¹²⁰ MIT, 2024

is still largely produced for defense. Any disruption in these supply chains could delay multi-billion-dollar projects and undermine investor confidence at a pivotal stage.

Capital Intensity and Financing Risk. FOAK cost estimates range from ~\$1B/GWe for pulsed FRC and Z-pinch systems to ~\$4B/GWe for laser ICF, though these figures remain unvalidated outside company roadmaps. Until fusion demonstrates sustained $Q > 1$ operation, financing is heavily equity-driven — reliant on venture investors, corporate backers, and government cost-sharing. These sources are sensitive to capital-market cycles, interest rates, and policy continuity. Withdrawal of key incentives such as the Inflation Reduction Act's credits could push LCOEs above competitive thresholds.

Siting and Public Perception. Fusion avoids the long-lived radioactive waste associated with fission, but it still requires tritium handling and creates neutron-activated materials. Communities unfamiliar with the distinction between fusion and fission may oppose new facilities, delaying reviews and permits. Repurposing brownfield sites such as retired coal stations or existing research campuses, with grid ties, water rights, and radiation licenses already in place, can reduce this risk substantially.

Competitive Technologies. Fusion competes with other firm clean-energy options, including SMRs, enhanced geothermal systems, long-duration storage, and natural gas with CCS. If these alternatives achieve materially lower capital costs or faster deployment, they could erode fusion's early-mover premium. At the same time, each competitor faces its own siting, financing, or scalability hurdles, giving fusion an opportunity to differentiate on energy density and long-term scalability.

Summary. Fusion's risk profile is defined by five critical areas: physics validation, supply-chain robustness, financing pathways, public acceptance, and relative competitiveness. Addressing these challenges early and decisively is essential to maintain momentum and credibility on the path to grid-connected deployment.

6. Conclusion and Recommendations

Fusion is no longer just a physics experiment — it's moving onto an investable track. The biggest opportunities in the near term sit across five themes: scaling up high-temperature superconductor (HTS) magnets, building reliable tritium breeding and handling systems, driving laser-diode efficiency into the 20–30% range, rolling out AI-driven plasma control to cut commissioning time and extend component life, and backing early project vehicles that have already secured long-term power purchase agreements (PPAs). Putting capital to work in these areas doesn't just back one design; it gives exposure to the whole sector, since every fusion pathway depends on at least one of these levers.

What to Watch. Over the next decade, several milestones will determine whether fusion truly crosses from promise to bankable reality. Commonwealth Fusion Systems aims for first plasma with its SPARC device in 2026, while Zap Energy is targeting engineering breakeven on FuZE-Q around the same time. Helion has committed to delivering megawatt-hours under its Microsoft PPA by 2028, a landmark moment if successful. General Fusion's liquid-metal demonstrator at Culham is expected to validate net-energy–relevant performance before 2030, while Xcimer is working toward pushing krypton fluoride laser efficiency beyond 20% by 2029. Finally, the U.S. Nuclear Regulatory Commission is expected to issue its first fusion operating license under Part 30 by the end of the decade. Each of these milestones, if met, will de-risk not only the companies involved but also the supply chains and enabling technologies that underpin the sector.

Next Steps for Investors. To get ahead of the curve, investors should:

- Map today's capacity and tomorrow's build-out plans for REBCO tape, lithium-6 enrichment, and high-power laser diodes, and start conversations with vendors open to offtake or equity deals.
- Run deep technical diligence on performance metrics — Q, pulse rates, component durability, and tritium cycles.
- Stress-test the numbers against possible policy shifts, from an early sunset of U.S. §45Y/§48E tax credits to tighter strike-price regimes in the U.K. and EU.
- Talk to offtakers — data centers, hydrogen producers, steelmakers — to pin down what PPA terms they'll actually sign.
- Audit brownfield and lab sites for grid connections, licensing feasibility, and community sentiment.
- Be clear on exit routes — IPOs, M&A, or yieldcos — while recognizing that fusion's timelines may be longer than early renewables.

Getting this work done over the next 6–12 months will give investors the conviction to move when the first commercial opportunities open up. Fusion sits at the rare intersection of financial upside and climate impact — and that makes it one of the most promising long-term bets in the global energy transition.

References

- Advanced Research Projects Agency-Energy (ARPA-E). (n.d.). BETHE: Breakthroughs enabling thermonuclear-fusion energy. <https://arpaenergy.gov/technologies/programs/bethe>
- Breakthrough Energy Ventures. (n.d.). *Portfolio: Commonwealth Fusion Systems*. PR Newswire. <https://www.prnewswire.com/news-releases>
- Clean Air Task Force. (2025, April). *Fusion on the grid – A U.S. fusion policy report*. <https://www.catf.us>
- Commonwealth Fusion Systems. (2021, November 30). *CFS closes US\$1.8 billion Series B*. CFS Energy. <https://cfs.energy>
- Congruent Ventures. (n.d.). *Avalanche Energy*. <https://www.congruentvc.com>
- European Commission. (n.d.). *ITER funding factsheet: EU contribution 2021–2027*. Energy. <https://energy.ec.europa.eu>
- Fusion Industry Association. (2024). *2024 global fusion industry report*. Fusion Industry Association. <https://fusionindustryassociation.org>
- Helion Energy. (2023, May 10). *Helion announces world's first fusion PPA with Microsoft*. Helion. <https://www.helionenergy.com>
- International Atomic Energy Agency. (n.d.). Fusion energy. <https://www.iaea.org/topics/fusion-energy>
- MIT Energy Initiative. (2024, September). *The role of fusion in a decarbonized electricity system*. <https://energy.mit.edu>

Resources for the Future. (2025, March). *Projected impacts of repealing §45Y and §48E technology-neutral clean electricity tax credits*. <https://www.rff.org>

U.S. Nuclear Regulatory Commission. (2023, February). *Options for licensing and regulating fusion energy systems* (10 CFR Part 30 pathway). NRC. <https://www.nrc.gov>

UK Government. (2025, June 12). *£2.5 billion for world-first prototype fusion energy plant*. GOV.UK. <https://www.gov.uk>